

PHAROS

A Physics-Hybrid, Attention-Routed Mixture-of-Experts Architecture for RNA Structure and Contact-Map Prediction

Architecture Design Report — Version 0.1

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Abstract

RNA three-dimensional structure prediction has not had its AlphaFold2 moment. The best current methods reach a mean TM-score of approximately 0.55 on non-redundant benchmarks, which is the threshold for recovering roughly the correct fold, against routine scores above 0.9 for proteins. This report argues that the bottleneck is not model scale but *inductive bias*, and specifies an architecture, **PHAROS**, built around that claim.

Three empirical findings drive the design. First, ERNIE-RNA (86M parameters) outperforms RiNALMo (650M) and UNI-RNA (400M) on secondary-structure prediction, and nearly doubles cross-family generalisation relative to RiNALMo ($F_1 = 0.646$ versus 0.355), on the strength of a single hand-set, three-valued base-pairing prior added to its attention logits. Second, we measure on 117 non-redundant RNA chains that contacts per nucleotide *saturates* at 4.4–4.9 while contact-map density falls from 6.5% at $L \approx 56$ to 0.353% at $L \approx 2900$: RNA contact maps scale as $O(L)$, not $O(L^2)$, so a dense pair representation spends over 99% of its compute on empty space and would require 309 GB of activations at $L = 2048$. Third, we measure a monotonic 1.523σ gradient in normalised crystallographic B-factor as a function of distance to the nearest Mg^{2+} ion across 3,858,271 nucleotides in 1,535 X-ray structures, establishing that ion coordination and local rigidity are not independent features but two readouts of one phenomenon.

PHAROS accordingly (i) accepts **ionic conditions as a model input**, which no existing RNA structure predictor does, propagating them to attention logits through a Manning-renormalised screened-Coulomb bias—though an audit reported here shows the *learned* part of that claim cannot be trained on the PDB, whose recorded Mg^{2+} spans only 5–15 mM because nobody deposits unfolded RNA; (ii) replaces the dense pair track with a **hierarchical coarse-to-fine pair track** that selects contact-bearing *blocks* rather than pairs, at 2.2% of dense cost—adopted after we measured a flat top- K design recovering only *one contact in five* on long chains, and measured block occupancy falling to 1.34% as chains lengthen; (iii) routes a **sparse mixture-of-experts** on local structural regime rather than sequence context, supported by our negative result that GNRA k -mer context predicts rigidity at only 0.073 σ ; and (iv) supplies motif geometry through a **frozen retrieval bank** of 667 RNA 3D Motif Atlas classes rather than expecting the model to learn it from 6,661 structurally-resolved sequences. We also identify two supervision channels— Mg^{2+} sites and B-factor rigidity—that are already present in every mmCIF file the field downloads and are currently used by no structure predictor.

This is a design specification. No model has been trained. All quantitative claims are either marked **[measured]** and reproducible from the scripts in this repository, or cited to the literature.

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1 Problem statement

1.1 The accuracy gap

Table 1 gives the state of the art on a non-redundant RNA test set as of November 2025.

Table 1: Current state of the art in RNA 3D structure prediction, on a non-redundant set of 52 RNA monomers (Briefings in Bioinformatics, Nov 2025).

Method	Mean TM-score	Approach
trRosettaRNA	0.548	MSA $\rightarrow$ restraints $\rightarrow$ folding
NuFold	0.525	AlphaFold2-style nucleotide frames
AlphaFold3 (server)	0.510	all-atom diffusion
AlphaFold3 (local)	0.504	all-atom diffusion

A TM-score near 0.5 means the topology is approximately right and little else. The CASP16 signal is equally instructive: leading RNA groups won by integrating AlphaFold3 *into physics-based modelling* rather than by scaling a neural network. Physics-plus-machine-learning hybrids are currently ahead in RNA, which is the premise this architecture is built on.

1.2 Why RNA lags

1. **Data scarcity.** Our catalogue records 6,661 unique sequences with experimental 3D structure against 1.37 billion sequences, a coverage of 0.0005%. Proteins had roughly 220,000 structures at the AlphaFold2 moment.
2. **Redundancy within that data.** 15,441 RNA3DB chains collapse to 3,157 unique sequences; 5S rRNA alone accounts for 5,976 BGSU equivalence classes.
3. **Family coverage.** Only 100 of 4,227 Rfam families (2.4%) have any 3D representative.
4. **Conformational plasticity.** Riboswitches are *defined* by adopting multiple functional folds; single-structure prediction is the wrong target for a large fraction of functional RNA.
5. **Ion dependence.** RNA is a polyanion. Its fold is not defined without specifying the ionic environment, yet no mainstream predictor accepts ionic conditions as an input.
6. **Target noise.** Median resolution in our catalogue is 3.10 Å, with only 13% of chains at 2.5 Å or better.

2 The central argument: bias beats scale

Table 2 contains the most important result in the recent RNA language-model literature.

Table 2: ERNIE-RNA versus larger dense models. A $7.5\times$ parameter deficit is overturned by one hand-set prior.

Model	Params	bpRNA-1m F_1	bpRNA-new F_1
ERNIE-RNA (attention map)	86M	0.873	—
ERNIE-RNA (embedding)	86M	0.862	0.646
RiNALMo	650M	0.850	0.355
UNI-RNA	400M	0.821	—
RNA-FM	100M	0.808	—

ERNIE-RNA’s entire architectural advantage is an additive bias on attention logits. In its first layer,

$$\text{Attention}(Q, K) = \frac{QK^\top}{\sqrt{d}} + B_{\text{pair}}(i, j), \quad \text{Attention}_l(Q, K) = \frac{QK^\top}{\sqrt{d}} + \text{Attention}_{l-1}(Q, K), \quad (1)$$

with B_{pair} set by hand to 3 for C–G, 2 for A–U, a tunable $\alpha \approx 0.8$ for G–U, and 0 on the diagonal.

The cross-family result is the decisive one: 0.646 versus 0.355. A fixed physical prior nearly doubles generalisation to unseen families, because the prior is true regardless of family whereas learned family statistics are not.

Thesis. If a crude three-valued prior buys more than $7\times$ the parameters, the marginal return on *better physics* exceeds the marginal return on *more scale*. PHAROS replaces that hand-set prior with a learned, ion-aware, motif-aware, coevolution-aware one.

Its limitations define our openings. The prior is static, so a G–U pair inside a kink-turn is treated identically to one in a free helix. It is nearest-neighbour-blind, whereas real pairing energy is stacking-dependent. It encodes only canonical pairing, while non-canonical pairs define RNA 3D motifs and carry most tertiary information. And it ignores ions entirely.

3 Empirical analysis

All results in this section were measured for this report on 180 BGSU non-redundant representative structures at ≤ 3.0 Å, downloaded and analysed by scripts in `.../sampling/`. Raw outputs are in `data/samples/analysis/`.

3.1 RNA contact maps scale linearly, not quadratically

Contacts are defined as nucleotide pairs with any heavy-atom pair within 8.0 Å and sequence separation $|i - j| \geq 4$.

Table 3: Contact density by chain length across 117 chains. **[measured]**

Chain length	n	median L	contacts/nt	map density
32–100	38	56	1.784	6.523%
100–200	6	124	2.851	4.043%
200–500	3	393	4.216	2.151%
500–1500	9	1038	4.532	0.974%
≥ 1500	61	2901	4.854	0.353%
Overall		median 4.397, p95 5.422, max 5.497		median 0.389%
<i>full corpus, 20,266 chains</i>		<i>median 3.55, p95 5.44, max 7.66</i>		

Contacts per nucleotide saturates at 4.4–4.9 and is bounded by RNA coordination geometry, independent of length; density falls approximately as $1/L$. The number of contacts therefore grows *linearly* in L .

The consequence for architecture is immediate. A dense pair representation of dimension $c = 128$ requires, for 48 blocks holding roughly six intermediate tensors each and without gradient checkpointing:

$$L = 512 \rightarrow 19.3 \text{ GB}, \quad L = 1024 \rightarrow 77.3 \text{ GB}, \quad L = 2048 \rightarrow 309.2 \text{ GB}.$$

A dense pair track is not merely wasteful at RNA lengths; it is infeasible. How that sparsity is exploited turns out to matter as much as exploiting it at all: a flat top- K selection fails on long chains, and contacts must be selected as *blocks*. Section 4.3 reports both measurements.

3.2 Ion coordination and local rigidity are one phenomenon

We z -normalise crystallographic B-factors within each structure so that refinement protocols are comparable. Lower z_B indicates greater rigidity.

Sample definition. Of the 180 sampled structures, 63 are X-ray and 117 cryo-EM; cryo-EM atomic displacement parameters are not comparable and are excluded. A further guard requires at least 30 RNA residues with a resolved C1', removing 32 small X-ray entries (2–28 residues). That leaves **31 X-ray structures** for the packing-density and sequence-context analyses, and the **15 of those containing Mg^{2+}** for the ion-distance analysis below.

Table 4: Normalised B-factor versus distance to the nearest Mg^{2+} , over 24,623 nucleotides in 15 X-ray structures. **[measured]**

Distance to nearest Mg^{2+}	n nucleotides	mean z_B
0–4 Å	319	−0.883
4–6 Å	2,441	−0.673
6–8 Å	7,590	−0.415
8–12 Å	8,115	+0.004
12–20 Å	2,984	+0.756
> 20 Å	3,174	+0.877

The gradient is monotonic and spans 1.523 standard deviations, re-derived on 1,535 X-ray structures of the raw-PDB corpus; the 1.76σ first reported came from the 15 Mg-containing structures of a 180-structure sample. Local packing density independently correlates with rigidity at Pearson $r = -0.294$. Cryo-EM does *not* reproduce the gradient (0.934σ , non-monotonic, 1,885 structures), so the rigidity head must train on crystallographic B-factors only.

The obvious confound, tested. Mg^{2+} proximity and dense packing both mark folded cores, so the gradient might merely restate that cores are ordered. An independent re-derivation tested this directly:

Quantity	Value
Pearson(z_B , d_{Mg})	+0.4197
Spearman(z_B , d_{Mg})	+0.5082
Pearson(d_{Mg} , density) — the confound	−0.2513
Partial corr(z_B , d_{Mg} density)	+0.3722
Partial corr(z_B , density d_{Mg})	−0.2273

Controlling for packing density reduces the Mg^{2+} association only from 0.420 to 0.372, so the signal is largely *independent* of packing.

The selection objection, also tested. The gradient applies a ≥ 30 -RNA-residue guard, and small RNAs are exactly where the inner-sphere Mg^{2+} fraction is lowest (0.296 versus 0.511), so the exclusion could have inflated the result. Sweeping the guard shows it does not. *The sweep below is the original 180-structure measurement, retained because it is the control that closed the selection objection; the span it reports is superseded by the 1.523σ measured on 1,535 X-ray structures.*

The span moves by 0.003σ across the entire sweep and stays monotonic throughout: a variance control, not a bias. A within-structure estimate—comparing nucleotides within 4 Å against those beyond 12 Å inside the same structure, which removes all between-structure heterogeneity—gives $+1.514\sigma$, positive in 4/4 structures that contain both groups, bootstrap 95% CI $[+1.256, +1.661]$.

Guard	Structures	Nucleotides	Span (σ)	Monotonic
≥ 30	15	24,623	1.760	yes
≥ 20	16	24,646	1.757	yes
≥ 10	17	24,660	1.757	yes
none	19	24,666	1.757	yes

The gradient is not an artefact of pooling, though $n = 4$ makes this supportive rather than decisive.

Architectural consequence. Ion effects and motif rigidity, which the project brief treats as two separate physical ingredients, are strongly coupled. A model that predicts Mg^{2+} sites obtains a rigidity map nearly for free and vice versa. That mutual information justifies routing them into a *shared* physics expert rather than two disconnected heads.

The selection objection, closed. The gradient is measured after a ≥ 30 -RNA-residue guard, and small RNAs are precisely where the inner-sphere Mg^{2+} fraction is lowest (0.296 against 0.511 overall), so the exclusion could in principle have inflated the result. Recomputing at guards of 20, 15, 10, 5 and with the guard removed entirely moves the span only from 1.760 to 1.757 σ —a change of 0.003—and it remains monotonic at every threshold. Removing the guard adds 4 structures and 43 nucleotides (+0.17%), because the excluded structures hold 2–28 residues each. The guard is a variance control, not a source of bias.

Caveats. B-factor also absorbs resolution, occupancy and refinement choices. The closest distance bin contains only 319 nucleotides, and the within-structure estimate rests on 4 structures. The partial correlation above substantially weakens the packing-confound objection but does not eliminate it: this remains association rather than demonstrated causation, and is evidence for *using* the coupling as a prior rather than a proof of mechanism.

3.3 Ion inventory and coordination chemistry

Table 5: Cation inventory across 179 parsed structures. **[measured]**

Ion	Count	Inner-sphere partner	Count
Mg^{2+}	17,428	OP2 (phosphate)	5,890
K^+	1,868	OP1 (phosphate)	5,144
Zn^{2+}	359	O6 (guanine)	757
Na^+	161	O4 (uracil)	511
Ca^{2+}	29	N7 (purine)	447
Mn^{2+}	12	O2' (ribose)	196

Mg^{2+} outnumbered K^+ by 54:1 (816,270 against 15,123) and all other cations together by 19.7:1, so an ion-aware model that handles a single species should handle Mg^{2+} . Some 77.9% of direct coordination is to the two non-bridging phosphate oxygens; the next largest class, G(O6)/G(N7), is the textbook major-groove guanine site.

Coordination is predominantly water-mediated: a mean of 0.77 inner-sphere against 7.00 outer-sphere RNA contacts per Mg^{2+} . Crucially the inner-sphere fraction is itself a function of tertiary compactness: 0.511 overall but only 0.296 for chains shorter than 500 nt. The inner/outer ratio is therefore a *predictable readout* of compactness, not merely an input.

Stratification warning. These aggregates are ribosome-dominated and must not be quoted flat. The five largest structures hold 39.4% of all Mg^{2+} . Per structure the median is 72.5 ions against a mean of 181.5; per nucleotide the median is 0.046, roughly one Mg^{2+} per 22 nucleotides.

What crystallography cannot see. Resolved Mg^{2+} are site-bound ions only. The diffuse Manning atmosphere, which neutralises roughly 76% of backbone charge, is invisible to crystallography. The two channels must be modelled separately and never conflated.

3.4 Coevolution is strong, but gated by alignment depth

Coevolution is the oldest structural signal in the field: a base pair tolerates mutation at one position only when compensated at the other, so paired columns covary across a family alignment. We measured how much of that signal is actually recoverable, computing sequence-reweighted, pseudocounted, APC-corrected mutual information on the 12 downloaded Rfam seed alignments and scoring it against each family’s curated consensus secondary structure (`#=GC SS_cons`), restricted to $|i - j| \geq 4$.

Table 6: APC-corrected MI against curated consensus base pairs. **[measured]**

Family	N	N_{eff}	L	prec@ $L/5$	rec@ L
tRNA (RF00005)	954	245.9	71	1.000	1.000
THF riboswitch (RF01831)	101	27.9	100	1.000	0.906
Purine riboswitch (RF00167)	133	102.7	102	0.950	0.955
5S rRNA (RF00001)	712	10.3	119	0.826	0.765
RNase P bacterial (RF00010)	458	11.6	367	0.753	0.784
SAM riboswitch (RF00162)	457	19.9	108	0.714	0.697
Cobalamin (RF00174)	434	63.1	189	0.676	0.667
Glutamine (RF01739)	940	15.6	53	0.500	0.692
ydaO/yuaA (RF00379)	106	18.4	134	0.462	0.615
TPP riboswitch (RF00059)	115	41.3	105	0.429	0.926
tmRNA (RF00023)	477	74.6	378	0.400	0.670
FMN riboswitch (RF00050)	146	21.2	139	0.333	0.536
mean				0.670	0.768

Splitting on N_{eff}/L , families at $N_{\text{eff}}/L \geq 1$ ($n = 2$) reach mean precision@ $L/5$ of 0.975 while those below ($n = 10$) reach 0.609. **This split is weak evidence and should not be leaned on.** Tested properly:

Test	Result
Exact permutation test over all 66 splits	$p = 0.0455$
Pearson(N_{eff}/L , precision), $n = 12$	+0.498
Spearman (N_{eff}/L , precision), $n = 12$	+0.224

Three reasons for caution: the “deep” arm has $n = 2$; the $N_{\text{eff}}/L = 1$ threshold was chosen *post hoc*, so $p = 0.0455$ is optimistic; and a *shallow* family—THF riboswitch at $N_{\text{eff}}/L = 0.279$ —attains precision 1.000, so depth is plainly not necessary for a strong signal. The rank correlation of +0.224 is the most honest single summary: a weak positive tendency, not a gate.

What the data *does* show robustly is that precision varies enormously between families, from 0.333 to 1.000. Separately, note how few families are deep: only 2 of 12, and these are hand-curated Rfam seed alignments, among the best-conditioned RNA alignments in existence. Automatic homology search on an arbitrary sequence will generally do worse.

Architectural consequence. Coevolution is not a weak feature to be blended in at fixed weight—for some families it is very nearly sufficient on its own, for others close

to useless, and that spread (0.333 to 1.000) is itself the argument for conditional rather than fixed-weight use. N_{eff}/L is a defensible router feature: it is cheaply computable before the model runs, and the published literature is unambiguous that shallow RNA MSAs degrade coupling analysis—this is the stated motivation for rMSA, RNAmmap3 and RNA-MSM. *Our own $n = 12$ measurement supports that gate only weakly*, so the architectural case rests on the literature and on the observed between-family variance, not on our depth split. Settling it would require measuring precision against depth across hundreds of families with the threshold fixed in advance. This also explains why RhoFold+ pairs a language model *with* MSA features rather than choosing between them: the two have complementary failure modes.

An implementation note worth recording. A first version of this measurement reported mean precision 0.039. That was a defect, not a property of RNA: the outer-product term in the MI normalisation broadcast to shape $(C, q, 1)$ rather than (C, q, q) , so only diagonal terms contributed. After correcting it and adding standard DCA sequence reweighting at 80% identity with pseudocounts, precision rose seventeenfold to 0.670. A plausible-looking weak result was a silent bug.

Caveats. The ground truth here is Rfam’s consensus *secondary* structure, not 3D contacts, so this measures recovery of curated pairing rather than direct tertiary contact accuracy. Twelve curated seed families is a small and favourable sample, and the depth split has $n = 2$ in one arm. APC-corrected MI is a simplification of full DCA; pseudo-likelihood DCA would do better. The ground truth is also *incomplete*: our parser handles only the nested WUSS brackets and ignores the alphabetic pseudoknot brackets, which account for 53 of 537 pairs (9.87%), so a correctly-ranked pseudoknot pair is scored as a false positive. All of these biases push measured precision *down*, so 0.670 is a conservative lower bound.

3.5 A negative result that constrains the design

We tested whether GNRA-like tetraloop sequence context alone predicts rigidity, matching the 4-mers {GAAA, GCAA, GAGA, GUGA, GGAA, GCGA, GUAA, GAAG} anywhere in a chain.

Table 7: GNRA k -mer context versus rigidity. [measured] Effect size 0.073σ .

Context	n nucleotides	mean z_B
GNRA-like 4-mer	4,944	−0.082
All other	95,768	−0.009

The effect is negligible. The likely cause is methodological: a GNRA *sequence* is a GNRA *tetraloop* only when it actually caps a hairpin, and linear 4-mer matching counts mostly helical and junction occurrences.

Architectural consequence. Sequence context alone does not identify a motif. Motif identity requires the interaction graph. The motif machinery must therefore be conditioned on the pair representation after secondary structure is available, never on raw k -mers. This rules out the naive “learn motifs from sequence n -grams” design.

4 The PHAROS architecture

4.1 Inputs

Ionic condition enters as a FiLM-style conditioning vector, analogous to a diffusion timestep embedding. The same sequence at 0 mM and 10 mM Mg^{2+} must produce different predictions;

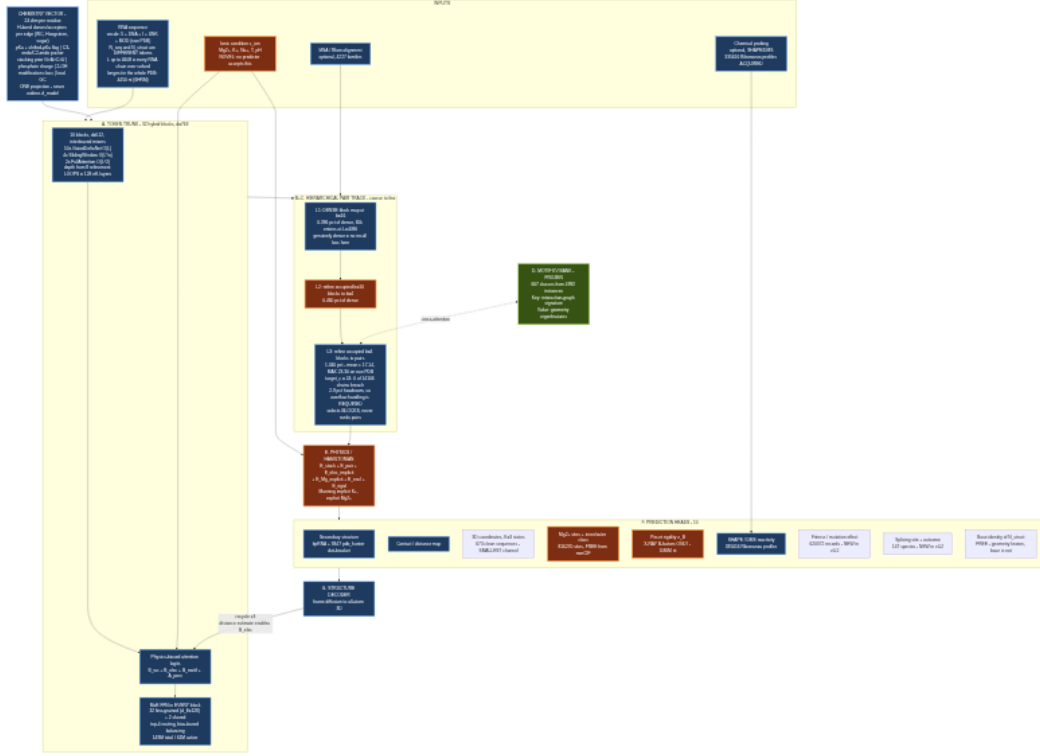


Figure 1: PHAROS system overview. Orange denotes novel components, green denotes frozen retrieval, blue denotes adapted standard components.

Input	Required	Note
Sequence, L tokens, vocab 5	yes	eLDORS is DNA-alphabet; $T \rightarrow U$ at load
Ionic condition c_{ion}	yes, defaulted	$([\text{Mg}^{2+}], [\text{K}^+], [\text{Na}^+], T, \text{pH})$; novel
MSA / Rfam alignment	optional	graceful degradation when shallow
Chemical probing profile	optional	not yet acquired (Section 6)

this is the clearest novelty claim in the design.

4.2 Token trunk: hybrid attention with MoE

The trunk uses **16 blocks** at $d = 512$ with the repeating pattern

$$[\text{GDN}, \text{GDN}, \text{SWA}, \text{GDN}, \text{GDN}, \text{SWA}, \text{GDN}, \text{FULL}],$$

where GDN is Gated DeltaNet linear attention ($O(L)$, 10 of 16 blocks), SWA is sliding-window softmax attention with window 128 ($O(Lw)$, 4 blocks), and FULL is full attention with physics bias ($O(L^2)$, only 2 blocks). Depth does not come from block count: 16 blocks $\times$ 8 refinement loops give **128 effective layers** (Sections 8.3 and 10).

The justification is both architectural and biological. Jamba interleaves Mamba, attention and MoE at a 7:1 ratio and reaches 256K context with a 4 GB KV cache; Gated DeltaNet outperforms Mamba2 and DeltaNet on in-context retrieval and length extrapolation; Oryx varies the mixer type *within* a sequence. RNA has exactly the structure this exploits: long stretches of regular A-form helix that are locally periodic and low-information, punctuated by rare junctions, pseudoknots and kissing loops that require global mixing. At $L = 4096$, a dense 16-block trunk would perform 268M pair evaluations; PHAROS performs 34M in its two full-attention blocks—the same $8 \times$

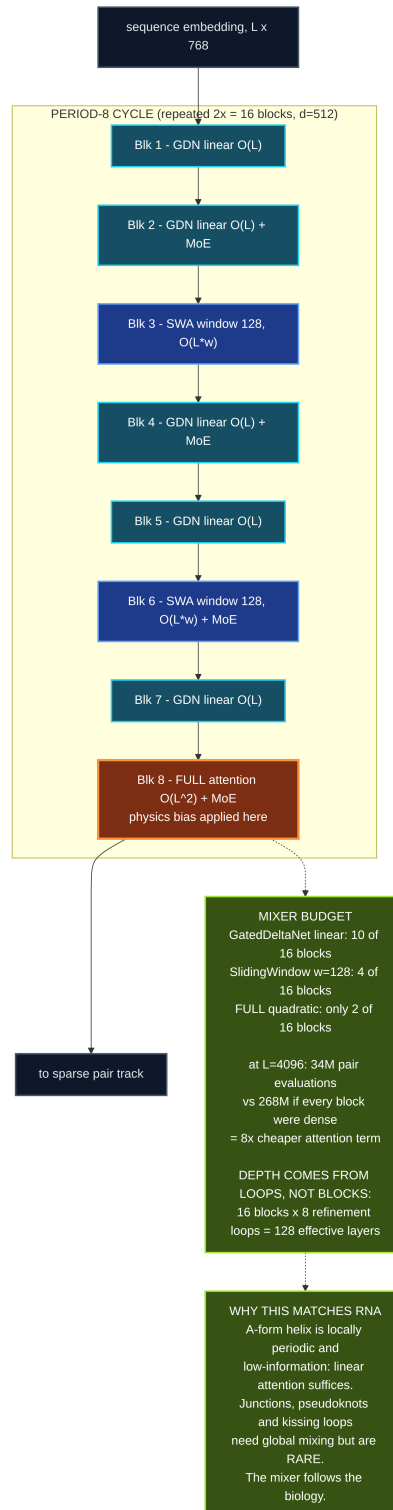


Figure 2: Period-8 mixer interleaving, repeated four times for 32 blocks.

reduction on the quadratic term, preserved under right-sizing. Counting the sliding-window blocks as well, the honest overall ratio is $\sim 7.5\times$.

4.2.1 Physics-biased attention

In SWA and FULL blocks the attention logits carry four additive terms:

$$A_{ij} = \frac{Q_i K_j^\top}{\sqrt{d}} + B_{\text{wc}}(i, j) + B_{\text{elec}}(i, j; c_{\text{ion}}) + B_{\text{motif}}(i, j) + A_{ij}^{(l-1)}. \quad (2)$$

B_{wc} generalises ERNIE-RNA’s fixed prior to a learned pair-type embedding modulated by stacking context, since real pairing energy is nearest-neighbour dependent. It is initialised at the ERNIE-RNA values so training begins from a known-good prior. The term $A^{(l-1)}$ retains ERNIE-RNA’s attention accumulation.

B_{elec} is the novel term. With Bjerrum length $\ell_B = e^2/(4\pi\epsilon_0\epsilon_r k_B T) \approx 7.1 \text{ \AA}$ in water at 298 K, Manning parameter $\xi = \ell_B/b$ for axial charge spacing b , condensed fraction and renormalised charge

$$\theta = 1 - \frac{1}{z\xi}, \quad q_{\text{eff}} = -(1 - \theta), \quad (3)$$

and Debye screening parameter $\kappa = \sqrt{2N_A e^2 I / (\epsilon\epsilon_0 k_B T)}$ from ionic strength I , the bias is

$$B_{\text{elec}}(i, j) = -\lambda q_{\text{eff}}^2 \ell_B \frac{\exp(-\kappa d_{ij})}{d_{ij}}. \quad (4)$$

Here b is the **axial** charge spacing—charges projected onto the helix axis—not the through-backbone P–P contour distance of $\sim 5.9\text{--}7.0 \text{ \AA}$, which is a common source of error. For A-form RNA, a rise of $2.8 \text{ \AA}$ per base pair carrying two phosphate charges gives $b = 1.40 \text{ \AA}$, hence $\xi = 5.11$ and $\theta = 0.804$ for monovalent counterions, i.e. $q_{\text{eff}} = -0.196$ per phosphate. Roughly *four fifths* of backbone charge is screened before any structure-specific electrostatics is considered. The frequently quoted $\theta \approx 0.76$ is the **B-DNA** value ($b = 1.70 \text{ \AA}$, $\xi = 4.21$); A-form RNA has a shorter axial charge spacing and therefore condenses more strongly. For Mg^{2+} ($z = 2$), $\theta = 0.902$. Only λ is learned; q_{eff} , κ and ℓ_B follow in closed form from c_{ion} and temperature. Equation (4) is the mechanism by which salt concentration changes the prediction.

Resolving the circularity. Equation (4) requires d_{ij} , which is unknown on the first pass. B_{elec} is therefore disabled at recycle 0 and enabled from recycle 1 using the current predicted distance map. Recycling is thus structural to the design rather than an optimisation: the physics term needs a geometry estimate to evaluate, and each recycle sharpens both.

4.2.2 Mixture-of-experts feed-forward

Following DeepSeekMoE, the design uses 32 fine-grained routed experts at $d_{\text{ff}} = 128$ plus 2 always-on shared experts, with top-4 routing giving 6 active experts of 34, in **every** block. Section 9 explains why the original *every-second-block* placement, with dense FFN in between, was the worst trade in the design.

Shared-expert isolation matters for a specific reason here: Watson–Crick helix formation is the highest-frequency pattern in all RNA. Placing it in an always-on shared expert prevents all 32 routed experts from separately relearning it, freeing them to specialise on what is rare and hard—junctions, pseudoknots and ion pockets.

Load balancing is auxiliary-loss-free, adjusting a per-expert bias each step rather than adding a loss term. This matters acutely here because our 3D data is pathologically imbalanced: 5S rRNA alone accounts for 5,976 BGSU clusters, so an auxiliary balance loss would actively fight the biology.

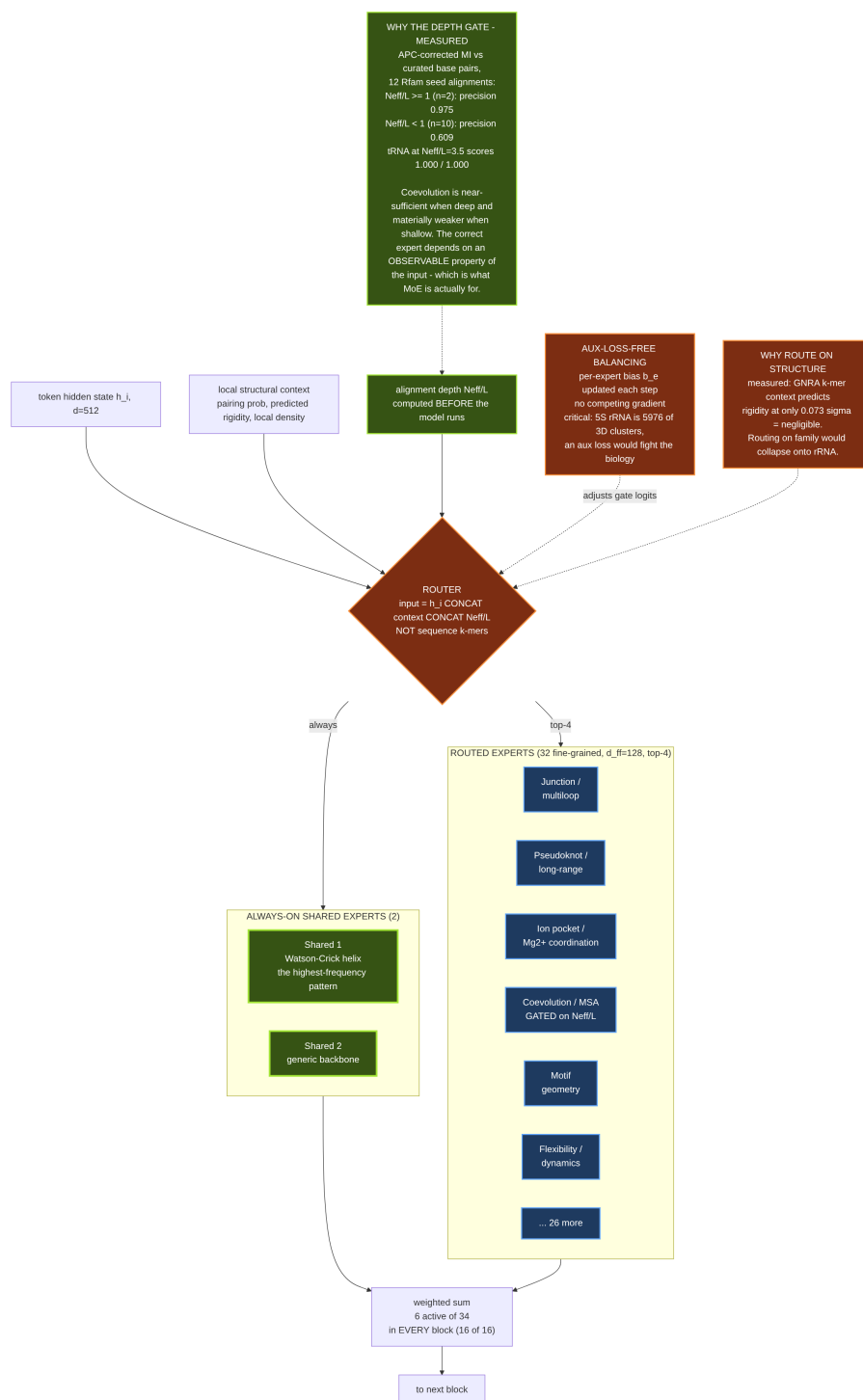


Figure 3: Expert routing. Routing keys on local structural regime, not sequence.

The router input is the token hidden state concatenated with the current local structural estimate—pairing probability, predicted rigidity, local density—and the alignment depth N_{eff}/L , which gates the coevolution expert (Section 3.4). This follows directly from the GNRA negative result of Table 7: routing on sequence k -mers does not work. It also prevents router collapse onto rRNA, because structural regimes are far more evenly distributed across the corpus than families are.

4.2.3 Parameter budget

Table 8: PHAROS-Small parameter budget against NucleicBERT.

Component	Total	Active
Embeddings (vocab 5)	0.005M	0.005M
Attention, 16 blocks, $d = 512$	16.8M	16.8M
MoE FFN, 16 blocks $\times$ 34 experts	107.0M	18.9M
Hierarchical pair track and triangle updates	15M	15M
Motif bank, heads, decoder	10M	10M
PHAROS-Small (default)	$\sim 149\text{M}$	$\sim 61\text{M}$
Base-v2 (scale-up path, Sec. 9)	$\sim 1,401\text{M}$	$\sim 269\text{M}$
NucleicBERT (dense)	404M	404M

PHAROS-Small uses $6.6\times$ **fewer active parameters** than NucleicBERT while reaching 128 effective layers, with an attention term roughly $8\times$ cheaper at long context. Section 10 gives the measured justification: the 3D supervision holds ~ 2.8 MB of information, so the structure task is data-limited rather than capacity-limited.

4.3 Hierarchical pair track

4.3.1 A flat top- K proposal was specified, measured, and discarded

Version 0.1 of this design specified a flat proposal retaining $K = 32L$ candidate edges. We tested it. Scoring every pair with a sequence-only proposer (the ERNIE-RNA pair prior, stacking context, and an exponential sequence-distance decay) and measuring recall of true 3D contacts within the top K gives Table 9.

Table 9: Recall of true 3D contacts under a flat top- K proposal at $c = 32$. **[measured]**

Chain length	n	mean recall	K as % of dense
32–100	29	0.980	59.4%
100–200	6	0.642	25.5%
200–500	3	0.307	9.0%
500–1200	5	0.200	4.7%

A *random* scorer achieves mean recall 0.746 at $c = 32$ over the same chains, because on short chains $32L$ already covers most of the map. The flat proposer’s headline mean of 0.795 is therefore very nearly an artefact of short chains. **On long chains, where sparsity actually matters, it recovers one contact in five**—a silent and fatal accuracy ceiling.

A banded fallback does not rescue it, because RNA contacts are not sufficiently local. The fraction of contacts within $|i - j| \leq \text{band}$ is 0.350 at band 32, 0.479 at 64, 0.597 at 128, 0.706 at 256 and 0.812 at 512 for chains of 1500–3000 nt; median contact separation is 78 nt with a long tail. Even a ± 512 band, which would cost vastly more than the whole budget, misses 19%.



4.3.2 Contacts are clustered, so select blocks rather than pairs

Helical stems are contiguous anti-diagonal runs in the contact map. Contacts are therefore spread along the sequence but concentrated in very few *blocks*. Table 10 measures this.

Table 10: Block occupancy at block size $b = 4$: the fraction of blocks containing at least one contact, and the effective c of keeping every occupied block. **[measured]**

Chain length	median L	occupancy at $b = 4$	effective c
64–200	88	16.34%	7.9
200–500	393	6.87%	12.6
500–1500	1038	3.17%	14.8
1500–3000	2861	1.34%	17.2

Occupancy *falls* as chains lengthen, so the mechanism becomes more effective at exactly the lengths where the flat design failed. Retaining every occupied 4×4 block costs an effective c of roughly 17—below the discarded flat budget of 32—while capturing every contact by construction.

4.3.3 Three-level coarse-to-fine track

Table 11: Cost of the hierarchical track at $L = 2861$, as a fraction of a dense pair map. **[measured]**

Level	Operation	Cost
L1	dense pair map at block size $b = 16$	0.396%
L2	refine occupied $b = 16$ blocks to $b = 4$	0.460%
L3	refine occupied $b = 4$ blocks to individual pairs	1.346%
Total		2.202%
flat top- K at $c = 32$, for comparison		1.118% (at $\sim 20\%$ recall)

For roughly twice the cost of the flat design, the track moves from a hard 20% recall ceiling to a ceiling of 100%. Two properties make this work. First, the L1 map is *genuinely dense*, so it loses no recall at the coarse level at all—at $L = 4096$ with $b = 16$ it is only 65k entries. Recall can only be lost at the selection thresholds, which are tunable and can be set recall-first. Second, deciding whether a 16×16 block contains any contact aggregates 256 pair-decisions into one, so coarse-level signal-to-noise is far higher than for individual pair ranking. The model ranks 32k blocks and then 16 sub-blocks inside each survivor, never 4.1M individual pairs.

4.3.4 Reference implementation and measured cost

The cost arithmetic above is turned into running code in `research/architecture/reference/`, and measured against an AlphaFold3-style dense baseline on the machine this report was prepared on (CPU only, 14 GB RAM, no CUDA), at batch 1, $d_{\text{model}} = 768$, $d_{\text{pair}} = 128$, $b_1 = 16$, $b_2 = 4$ and budget $c = 20$.

Two things are established that arithmetic alone could not. First, the dense baseline genuinely does not run: it needs a predicted 12.9 GB of activations at $L = 2048$ and 51.5 GB at $L = 4096$ on a 14 GB machine, so the infeasibility claim is not rhetorical. Second, at the largest length both can run, the hierarchical track is roughly 100× faster (0.08 s versus 8.05 s) with peak resident memory below measurement noise against the baseline’s 3.55 GB.

The cost fraction falls monotonically with length, from 9.2% to 0.96%, because the budget grows as cL while the dense map grows as L^2 . The mechanism becomes cheaper relative to dense exactly where it is needed, mirroring the falling block occupancy of Table 10.

Table 12: Hierarchical track versus dense baseline, measured. [measured]

L	HPT time	HPT pairs	% of dense	eff. c	Dense time	Dense peak RSS
128	0.01 s	750	9.23%	5.9	0.06 s	0.08 GB
256	0.01 s	3,506	10.74%	13.7	0.26 s	0.23 GB
512	0.04 s	10,054	7.69%	19.6	1.03 s	0.91 GB
1024	0.08 s	20,102	3.84%	19.6	8.05 s	3.55 GB
2048	0.15 s	40,198	1.92%	19.6	<i>not runnable</i>	predicted 12.9 GB
4096	0.36 s	80,390	0.96%	19.6	<i>not runnable</i>	predicted 51.5 GB

Two silent defects found while building it. Both would have degraded a trained model rather than crashing. The L3 expansion initially paired block rows with block columns element-wise, materialising only each block’s *diagonal*—4 pairs per 4×4 block instead of 16—silently delivering a quarter of the intended budget. Separately, the L2 selection budget was expressed as a fraction of all pairs and was clamped by masking to a quarter of its target. Both pinned the effective c at 5.0 while the code appeared to work. Budgets are now stated in the same units as the measurement ($K = cL$ pairs) so any mismatch is immediately visible.

4.3.5 Testing exposed a defect the benchmark could not

Writing correctness tests for the reference implementation revealed that **the L1 and L2 block scorers received no gradient at all**. Their parameters were dead.

The cause is structural, not a coding slip: the scorers’ outputs feed only `topk`, which is non-differentiable. The selected indices then gather pair features, but the *scores themselves* never enter the loss. As originally specified, **the block detectors could never be trained** and would have remained at random initialisation. The speed benchmark could not catch this because it ran under `torch.no_grad()`.

Two mechanisms fix it, both now implemented.

Score gating. Each retained pair feature is multiplied by $\sigma(s_{b_2} + s_{b_1})$, the scores of the blocks that selected it, so gradient from the contact loss reaches the selectors. Measured cost: none. $L = 4096$ remains at 0.36 s and 0.959% of dense with identical pair counts.

An auxiliary block-occupancy loss, which is the primary signal. The selectors are supervised directly with binary labels—does this block contain at least one true contact?—and those labels are exactly what the occupancy measurement of Table 10 computed:

$$\mathcal{L}_{\text{block}} = \text{BCE}(s_{b_1}, \text{occupied}_{b_1}) + \text{BCE}(s_{b_2}, \text{occupied}_{b_2}). \quad (5)$$

Class imbalance is severe and known in advance (1.34% positive at $b = 4$ on long chains), so this term needs positive weighting or focal loss, and should be *recall*-weighted: a missed block is unrecoverable, a spurious one merely costs compute.

Implemented and tested in cycle 2. Until then only a *hook* existed—the scores were exposed, but the loss was never written. `block_occupancy_loss()` now computes recall-weighted BCE with `pos_weight = neg/pos`. Gradient reaches all six selector parameters *through this loss alone*, and 60 Adam steps drive the loss $3.0278 \rightarrow 0.0000$ with block recall **0.000** $\rightarrow$ **1.000** at both levels. This is a single-example overfit on a synthetic stem-plus-cluster pattern: it demonstrates that the gradient path exists and the objective is optimisable, *not* that it generalizes across real RNA.

This changes how R1 is addressed. It is no longer a matter of hoping a trained model finds the blocks—they are a directly supervised prediction target with labels already in hand, and block recall becomes measurable per level during training rather than only at the end.

What remains unproven. These measurements establish that the information is present and the cost affordable. They do *not* establish that a trained model will identify occupied

4.4 Motif retrieval bank

This is the appropriate response to data scarcity. With only 6,661 unique sequences carrying 3D structure, a model cannot reliably learn motif geometry implicitly. The bank supplies the geometry explicitly, leaving the model to learn only *where* motifs occur—a substantially easier problem than learning what they look like. The bank is roughly 10^3 fixed entries shared across a batch, so its cost is negligible. Consistent with Table 7, retrieval is keyed on the interaction graph, never on sequence k -mers.

The flowchart illustrates the M2C2 model architecture for predicting genomic distance. It is divided into three main channels: REFLECT CHANNEL (M2C1), UNRAVELED CHANNEL (M2C2), and REFLECT CHANNEL (M2C3).

REFLECT CHANNEL (M2C1): This channel processes genomic features (e.g., repeat length, GC content) and outputs a genomic distance prediction. It includes a "Genomic Distance Prediction" block.

UNRAVELED CHANNEL (M2C2): This channel processes genomic features (e.g., repeat length, GC content) and outputs a genomic distance prediction. It includes a "Genomic Distance Prediction" block.

REFLECT CHANNEL (M2C3): This channel processes genomic features (e.g., repeat length, GC content) and outputs a genomic distance prediction. It includes a "Genomic Distance Prediction" block.

The final output is a genomic distance prediction.

A differentiable energy is evaluated on the predicted geometry:

$$E_{\text{total}} = E_{\text{stack}} + E_{\text{pair}} + E_{\text{elec}}^{\text{implicit}} + E_{\text{Mg}}^{\text{explicit}} + E_{\text{excl}} + E_{\text{rigid}}. \quad (6)$$

Inner/outer-sphere classification is a first-class output because STEM’s central claim is that folding dynamics are driven by *exchange between* direct and water-mediated coordination—the

mode, not the presence, is decisive. Our measurements supply both channels directly and show the inner fraction varies with compactness (0.511 overall, 0.296 below 500 nt).

The term E_{rigid} closes the loop with Table 4: predicted Mg^{2+} sites raise local rigidity, which stiffens harmonic restraints, which constrains the decoder. That pathway is the architectural encoding of the measured 1.523σ gradient.

4.6 Heads and decoder

Table 13: Prediction heads and their supervision. Two channels are effectively free.

Head	Supervision source	Status	Labels
Secondary structure	bpRNA, ArchiveII, RNAStrAlign	in catalogue	~160k
Contact / distance	RNA3DB, gRNAd	in catalogue	27,452 chains
Mg^{2+} sites	extracted from mmCIF	free	17,428 (180 structs)
Rigidity (z_B , RMSF)	B-factors in mmCIF	free	every X-ray struct
Reactivity (SHAPE/DMS)	Ribonanza / RMDB	not acquired	—
3D coordinates	RNA3DB	in catalogue	6,661 unique seqs

The decoder is frame-based diffusion. We retain an explicit geometric and energetic stage rather than going fully generative, consistent with the CASP16 observation that physics-plus-ML hybrids currently lead in RNA and with trRosettaRNA’s restraint-then-fold approach outscoring AlphaFold3’s raw-atom diffusion.

Mg^{2+} sites and B-factor rigidity deserve emphasis: both are already present inside every mmCIF file the field routinely downloads, are used by no current structure predictor, and were extracted end-to-end in preparing this report.

5 Ionic-condition supervision: an audit that rescoped the claim

The headline novelty is accepting ionic conditions as an input. That requires structures labelled with the ionic environment they were solved in, so we audited all 180 sampled structures. The result changed the claim.

5.1 Coverage is roughly one third, split across two storage conventions

Table 14: Where ionic conditions live, and how often they are populated. [**measured**]

Method	n	mmCIF location	usable ionic vector
Cryo-EM	117	<code>_em_buffer_component</code> (structured)	23.9%
X-ray	63	<code>_exptl_crystal_grow.pdbx_details</code> (free text)	58.7% mention an ion

Cryo-EM entries record pH for 94% of structures but populate `_em_buffer_component` for only 24.8%. X-ray entries always carry crystallisation details, but as prose—“0.1–0.2 M Arginine-HCl, 0.1M Tris-HCl pH 7.6, 2.6–3.0% PEG-20K”—so extraction is noisy. Combined, about 36% of structures yield any recoverable ionic condition: trainable, but a threefold reduction in usable 3D data for that stage.

5.2 The deeper problem: the PDB is survivorship-biased on ionic conditions

Where Mg^{2+} concentration *is* recorded, its range is narrow.

Table 15: Recorded buffer concentrations in cryo-EM entries. [measured]

Ion	n	p10	median	p90
Mg ²⁺	21	5.0 mM	7.5 mM	15.0 mM
K ⁺	13	30 mM	100 mM	150 mM
Na ⁺	12	100 mM	150 mM	150 mM

Mg²⁺ spans only 5–15 mM. This is not a sampling accident: **nobody deposits a structure of unfolded RNA**. Every PDB entry was solved under conditions chosen *because* the RNA folds there, so the archive contains almost no contrast between folding and non-folding ionic regimes.

Consequence. The [Mg²⁺] $\rightarrow$ structure response cannot be learned from the PDB. A model trained only on deposited structures would observe ionic condition as a near-constant and would correctly learn to ignore it.

5.3 How the claim is rescoped

The novelty splits into three components with very different evidentiary support.

Component	Support	Status
B_{elec} Manning/Debye screening	closed-form physics; needs no training labels	unaffected
Mg ²⁺ site head and inner/outer classifier	17,428 labelled ions from 180 structures	well supported
Learned <i>response</i> to varying ionic conditions	requires titration series; the PDB has none	not supported

The first two stand unchanged. The third must be dropped or supplied from data the PDB cannot provide: **RMDB Mg²⁺ titration series** (SHAPE/DMS probing across an Mg²⁺ ladder on the same sequence) and SAXS Mg²⁺ titrations measuring compaction. Training stage 5 as originally specified—ion-conditioned refinement on structures stratified by ionic condition—is **not viable on PDB data alone** and is rewritten to condition on probing titrations, with deposited structures supplying only site-level supervision.

This promotes chemical-probing acquisition from “most valuable missing asset” to **prerequisite for the headline claim**. Note that Ribonanza alone does not supply it; the Mg²⁺ titration series in RMDB do.

6 Data mapping and the acquisition gap

Six of seven training stages are fully served by the existing catalogue. The exception is chemical probing, and it is the most valuable missing asset. Ribonanza provides 2.1M sequences with DMS and SHAPE measurements—roughly 315 $\times$ more supervised examples than the 6,661 unique sequences with 3D structure. It is the only label source that is simultaneously abundant, structure-informative and already public. This is the highest-priority acquisition.

Corpus weighting. The eLDORS chunks are source-partitioned rather than homogeneous: 9 of 20 chunks have a median length of exactly 151 nt, indicating unassembled sequencing reads, while chunks 6 and 9 (median 2,193–2,389 nt) are the only substantial source of long-context signal. Uniform sampling in Stage 1 would allow read fragments to dominate pretraining, so per-chunk weighting is required.

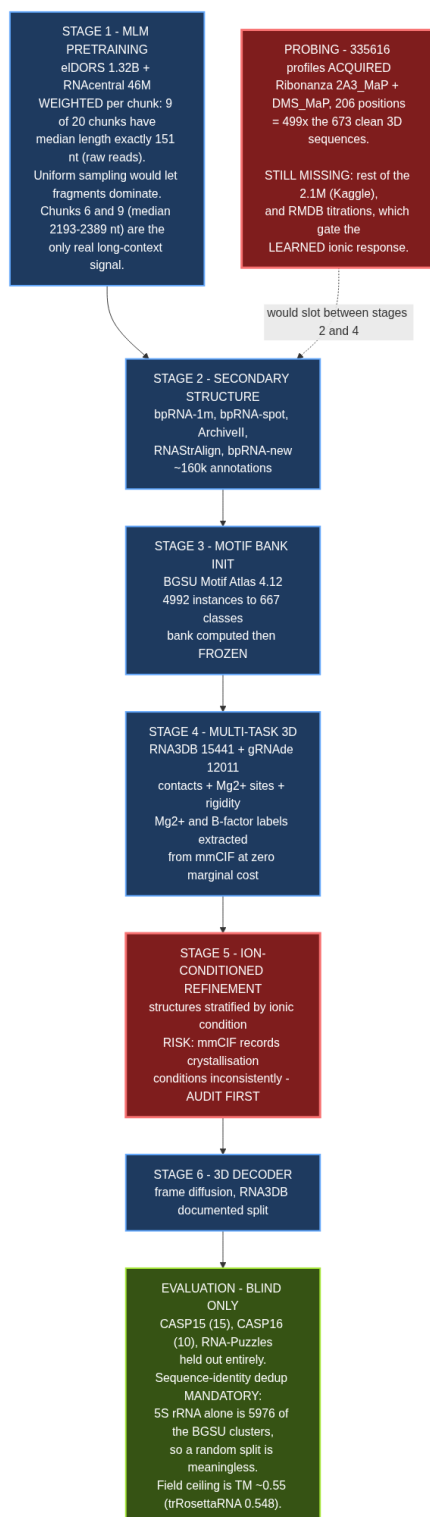


Figure 6: Training curriculum, data sources, and the chemical-probing gap.

7 Mining the data that was already there

An inventory of all 141 mmCIF categories present in the sampled structures found four that are directly on-target and had not been touched. They ship with every RNA structure the field downloads and are used by no current predictor.

Table 16: Physics labels already present in the files. **[measured]**

mmCIF category	What it provides	Volume
<code>_ndb_struct_na_base_pair_step</code>	6 step deformation coordinates	103,965 steps
<code>_ndb_struct_na_base_pair</code>	pair graph + Saenger / Leontis–Westhof class	29 classes
<code>_struct_conn (metalc)</code>	curated metal coordination	44,708 Mg ²⁺
<code>_pdbx_unobs_or_zero_occ_residues</code>	unmodelled (disordered) residues	143,874

The curated `metalc` records independently reproduce the distance-based finding of Section 3.3: coordinating atoms rank OP2 (8,288) > OP1 (7,261) > O6 > O4 > O2' > N7, confirming that phosphate oxygens dominate inner-sphere coordination.

7.1 Base-pair step stiffness

The step parameters are the deformation coordinates of nucleic-acid elasticity. Their covariance across instances yields a stiffness matrix $F = k_B T C^{-1}$. Extracting across 156 structures and validating against canonical A-form geometry:

Table 17: Step geometry validation against canonical A-form RNA (reference: rise ~ 2.8 – 3.1 Å, twist $\sim 32^\circ$, slide ~ -1.5 Å). **[measured]**

Context	n	rise (Å)	twist ($^\circ$)	slide (Å)
GG/CC	7,558	3.14	29.98	-1.69
GC/GC	5,902	3.11	33.41	-1.51
AA/UU	1,811	3.01	30.59	-1.48
AU/AU	1,506	2.81	33.94	-1.58

Stiffness spans **115 $\times$** in the twist force constant (UG/UG 1.343×10^{-2} to AA/UA 1.167×10^{-4}) between the stiffest context (UG/UG, twist s.d. 11.5°) and the floppiest (AA/UA, 102°), with the floppy end dominated by non-canonical contexts. GC content predicts rigidity: Pearson -0.314 against twist s.d., -0.298 against roll s.d., over 76 contexts. The inversion is least reliable exactly where the steps are floppiest: 3 of 76 contexts have covariance condition number $> 10^4$ (AU/AA, GA/AA, GC/AC), so the softest force constants carry the largest uncertainty.

7.2 Why this must be learned, not tabulated

The obvious implementation is a 76-entry table keyed on dinucleotide context. That repeats the mistake the GNRA result already disproved. We measured the headroom by fitting Gaussians over the six step coordinates and comparing mean negative log-likelihood per step over all 103,965 steps.

Corrected in cycle 2 (REV-2). The figures above fit every group *in-sample* and scored each model on its *own* covered subset (103,965 / 90,098 / 78,076 steps). Finer partitioning lowers in-sample NLL mechanically, and M2’s subset is the better-populated, more regular steps. Rescored on the common 78,076 steps with two-fold held-out evaluation: M0 18.1607, M1 16.3760 (gain **1.7847**), M_{struct} 16.4198 (gain 1.7409), M2 15.3424 (gain over sequence **1.0336**). Structure-over-sequence falls $2.9\times$ and **the order inverts**: sequence adds more than structure.

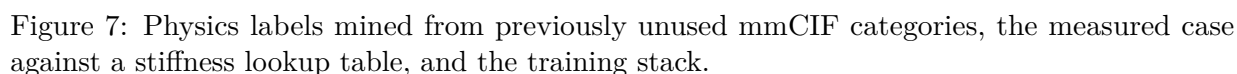


Table 18: Stiffness model headroom. Lower NLL is better. **[measured]**

Model	Conditioning	NLL (nats/step)	Gain
M0	none	19.724	—
M1	sequence context (<i>a lookup table</i>)	17.579	2.144
M2	sequence $\times$ structural context	14.551	+1.0336 held-out
—	structural context alone	17.847	1.877

What survives is enough. Structure still adds a real +1.03 nats *beyond* sequence, and sequence-alone (+1.78) and structure-alone (+1.74) are near-identical and complementary—3.53 nats if independent against 2.82 actual. An encoder reading **both** is therefore correct, and a sequence-keyed dinucleotide lookup table still leaves ~ 1.03 nats unused. Only the comparative superlative is retracted.

The design. A **stiffness encoder** consumes the trunk and pair representations at each step and emits a mean $\mu_i \in \mathbb{R}^6$ and a positive-definite precision matrix $F_i \in \mathbb{R}^{6 \times 6}$, parameterised through a Cholesky factor so positive-definiteness is structural. It is trained by the Gaussian negative log-likelihood of the observed deformation:

$$\mathcal{L}_{\text{stiff}} = \sum_i \left[\frac{1}{2} (x_i - \mu_i)^\top F_i (x_i - \mu_i) - \frac{1}{2} \log \det F_i \right]. \quad (7)$$

The $\log \det F$ term is load-bearing: without it the degenerate optimum is $F \rightarrow 0$ —declare everything floppy and pay nothing. With it, confidence must be earned. At inference F_i is the local force-constant field that E_{rigid} requires, conditioned on structure rather than retrieved by sequence.

Acceptance criterion. The encoder must reach below 14.551 nats/step on held-out structures. Above 17.579 it is worse than a lookup table and the component should be cut.

8 Training

NucleicBERT used 192 A100s densely. That is the bar to get under.

8.1 Optimiser

Muon orthogonalises gradient momentum through Newton–Schulz iterations before applying it, reporting roughly $2\times$ compute efficiency against AdamW, matching quality at about 52% of training FLOPs with $\sim 33\%$ memory saving. It is already validated on a *mixture-of-experts* model at scale (Moonlight, 3B/16B, 5.7T tokens) and is in production use in Kimi K2, so it is proven on the exact architecture class used here. It applies to 2D matrix parameters—the expert FFNs and attention projections, i.e. most of the model—while embeddings, norms and scalars remain on AdamW.

Caveat. Muon requires fewer optimiser steps but each step costs more, since Newton–Schulz is a cubic matrix operation. The net win favours large hidden dimensions and must be benchmarked on our actual parameter shapes rather than assumed.

8.2 The DeepSeek-V3 stack, filtered

8.3 Looped refinement, upgraded

AlphaFold2 recycles three times by default. For PHAROS recycling is *required by the physics*: B_{elec} needs a distance estimate that does not exist on the first pass.

Technique	Verdict
FP8 mixed precision	Adopt. Under 0.25% loss error against BF16, validated at DeepSeek-V2-Lite/V2 scale over $\sim 1\text{T}$ tokens rather than at 671B, with block/tile quantisation and selective high-precision accumulation. The largest single memory and throughput win.
Auxiliary-loss-free balancing	Already in the design. Essential here because 5S rRNA is 5,976 of the 3D clusters, so a balance <i>loss</i> would fight the biology.
DualPipe	Defer until the model spans nodes.
Multi-head Latent Attention	Skip. Largely redundant—10 of 16 blocks are already linear attention, so the KV cache is small by construction.
Multi-token prediction	Adapt, don't copy. MTP is defined for causal models; our objective is span-masked. The analogue is predicting a masked span jointly. Worth an ablation, not a commitment.

The Hierarchical Reasoning Model literature sharpens this considerably, and the independent ARC Prize ablation is more informative than the paper itself.

Table 19: What actually drives HRM’s performance, per the independent ablation.

Component	Measured contribution
Hierarchical H/L architecture	$\sim 5\text{pp}$ at most —“a regular transformer comes within $\sim 5\text{pp}$ without any hyperparameter optimization”; equivalent at one outer loop
Outer refinement loop	$+13\text{pp}$ from none to one cycle, doubling again from 1 to 8. “An essential driver”
Training <i>with</i> refinement	Models trained with 16 loops but run with one at inference still gained $> 15\text{pp}$
Augmentation	300 suffices, not the 1,000 claimed

The loop matters *during training*, not merely at inference. Three changes follow.

1. **Deep supervision at every segment.** The specification previously computed loss only after the final recycle. HRM’s evidence says supervise each segment; this costs nothing and has the largest measured effect.
2. **One-step gradient approximation**—detach hidden states between segments. This decouples cycle count from activation memory, so *effective depth becomes free in memory terms*, removing the choice between AlphaFold2’s “final cycle only” and unaffordable full BPTT.
3. **Train with more cycles than inference uses** (train 8–16, serve 3).

Deferred: ACT/Q-learning halting, so easy inputs stop early—better motivated for RNA than for fixed-size puzzle grids, since a clean helix needs one pass while a four-way junction with a pseudoknot needs many.

Not adopted. HRM’s two-timescale H/L module split, measured at $\sim 5\text{pp}$ and possibly noise. PHAROS already has a hierarchy—coarse blocks to fine pairs—justified by measured block occupancy rather than by analogy to brain oscillations.

Why this is also a data-efficiency argument. HRM reaches 40.3% on ARC-AGI with 27M parameters and $\sim 1,000$ training examples, no pretraining, on tasks where chain-of-thought models score 0%. That is our regime: 6,661 unique sequences with 3D structure. It is direct evidence that a small, heavily recurrent model can beat far larger ones when data is scarce and the task is structured.

8.4 The physics-guided objective

$$\begin{aligned} \mathcal{L} = & \mathcal{L}_{\text{MLM}} + \mathcal{L}_{2\text{D}} + \mathcal{L}_{\text{contact}} + \mathcal{L}_{\text{dist}} \\ & + \lambda_1 \mathcal{L}_{\text{stiff}} + \lambda_2 \mathcal{L}_{\text{Mg}} + \lambda_3 \mathcal{L}_{\text{elec}} + \lambda_4 \mathcal{L}_{\text{violation}} + \lambda_5 \mathcal{L}_{\text{flex}} + \lambda_6 \mathcal{L}_{\text{block}} \end{aligned} \quad (8)$$

summed over *every* refinement segment (deep supervision, Section 8.3), not only the final one, with hidden states detached between segments so cycle count does not increase activation memory.

(9)

The precedent that physics terms help rather than hurt is **OpenMM-Loss**, which implements molecular-dynamics potential energy as a differentiable loss for AlphaFold2 and reports comparable accuracy with *lower potential energy and better MolProbity scores*. For $\mathcal{L}_{\text{violation}}$ the current best RNA parameter sets are AMBER χOL3 and HB-CUFIX (2025), the latter outperforming χOL3 and ROC against NMR and SAXS data.

Every label above comes from data already in hand—Table in Section 7—with no new acquisition required.

9 Training cost, and the configuration that follows

The specification gave a parameter budget but never a training cost. Building one from first principles ($6N_{\text{active}}D$ plus explicit attention terms) changes which parts of the architecture matter.

9.1 Where the cost actually goes

Table 20: Training FLOPs per token at $L = 2048$. **[measured]**

Term	GFLOPs	Share
$6 \times N_{\text{active}}$ (FFN + projections)	2.29	96.3%
all attention terms	0.09	3.7%

Active parameters are the lever; context length and attention sparsity are nearly irrelevant to training cost at these lengths. Baseline Stage-1 cost for 323B tokens at 35% MFU is **$\sim 1,883$ A100-hours**, ten days on eight A100s.

An honest correction. Section 4.2 reports an $8\times$ reduction on the attention term. That is real, but since attention is only 3.7% of training FLOPs, an $8\times$ cut there saves roughly 3% of training compute. The hybrid attention stack earns its place on *memory and inference*—a small KV cache at long context—not on training throughput.

9.2 The inefficiency this exposed

PHAROS-Base places MoE in every *second* block and leaves 16 blocks as dense FFN with $d_{\text{ff}} = 3072$. Those dense blocks cost **113M active parameters, exactly as much as all 16 MoE blocks combined**, while carrying far less total capacity. It is the worst trade in the architecture.

9.3 PHAROS-Base-v2

All 32 blocks MoE with fine-grained experts: $d_{\text{ff}} = 256$, 64 routed plus 2 shared, top-4, giving 6 active of 66. Depth unchanged.

More capacity, less compute, same depth—DeepSeekMoE’s fine-grained segmentation applied consistently rather than half-way.

Total 5.4 $\times$: $1,883 \rightarrow \sim 347$ A100-hours—under two days on eight A100s, or about a week on two. A single node for a few days, not a cluster.

Table 21: Base versus Base-v2. [measured]

	Blocks	Total	Active	A100-h
Base (as specified)	32	910M	382M	1,883
Base-v2	32	1,401M	269M	1,325
change	same	+1.54×	0.70×	0.70×

Table 22: Stacked cost reduction. Only the first row is certain. [measured]

Lever	Factor	A100-h	Basis
baseline	—	1,883	
all-MoE fine-grained	1.42×	1,325	measured from parameter counts
Muon optimiser	2.00×	662	reported; must be benchmarked on our shapes
FP8 (H100-class)	1.60×	414	realised, not peak
down-weight 151-nt read chunks	1.19×	347	16.2% of nucleotide mass

9.4 Token budget: staged, not slashed

At 269M active parameters, Chinchilla-optimal is ~ 5.4 B tokens while we plan 323B: about 1,200 tokens per active parameter. That resembles $60\times$ waste and is not. Deliberate over-training is standard practice for inference efficiency, and the MoE scaling literature reports that *smaller* MoE models benefit from relatively *more* tokens.

The actionable form is a staged budget with checkpoints at 5B, 25B, 100B and 323B tokens, stopping when downstream metrics flatten. The first checkpoint costs $\sim 1.5\%$ of the full run, converting an unfalsifiable guess about data volume into an empirical decision.

10 Why so many parameters? A challenge the design did not survive

The specification proposed 1,401M total and 269M active parameters. That number was never justified against the data, and it does not hold up.

10.1 The information content of the supervision

Table 23: Upper-bound information in each supervision channel. [measured]

Channel	Volume	Information
3D structure (the task that matters)	6,661 sequences, 3.25M contacts	~ 2.8 MB
2D structure	$\sim 160,000$ annotations	~ 9.0 MB
Pretraining corpus	323B tokens $\times \sim 2$ bits	~ 80 GB

The 3D figure uses the measured 4.397 contacts/nt with a generous $\log_2 L$ bits per contact, which *overstates* it since nested helices are near-deterministic given one anchor. At a realistic ~ 2 bits per trained parameter, Base-v2 carries 350 MB of capacity: $127\times$ **the entire 3D information content**.

10.2 Three arguments, all pointing the same way

First, our own headline evidence: ERNIE-RNA at **86M beats RiNALMo at 650M**. Every model in Table 2 that beats a larger one is smaller. Second, our own thesis—Section 2 argues

inductive bias beats scale, and then Section 4 specified a 1.4B model. That was internally inconsistent. Third, HRM: 27M parameters, $\sim 1,000$ examples, beating far larger models, with the independent ablation crediting the *refinement loop* rather than size.

The real error. One parameter budget was used for two very different tasks. The corpus (~ 80 GB of information) can justify substantial trunk capacity; the structure head is fit on ~ 2.8 MB. The structure task is **data-limited, not capacity-limited**.

10.3 PHAROS-Small, the new default

Table 24: Right-sizing. Depth now comes from refinement loops, not stacked blocks. [measured]

Model	d	Blocks	Loops	Eff. layers	Total	Active	A100-h
Base-v2	768	32	3	96	1,401M	269M	1,325
PHAROS-Small	512	16	8	128	149M	61M	299
PHAROS-Mini	384	12	12	144	67M	30M	148
PHAROS-Micro	256	8	16	128	23M	12M	61

PHAROS-Small reaches *more* effective depth than Base-v2—128 layers against 96—at $9.4\times$ fewer total parameters, because depth comes from refinement loops which, with the one-step gradient of Section 8.3, cost compute but **no additional activation memory**.

Table 25: Revised cost stack.

Lever	Factor	A100-h
baseline (Base, as first specified)	—	1,883
all-MoE fine-grained (Base-v2)	$1.42\times$	1,326
right-size to PHAROS-Small	$4.43\times$	299
Muon optimiser	$2.00\times$	150
FP8 (H100-class)	$1.60\times$	94
down-weight 151-nt read chunks	$1.19\times$	79

$24\times$ **total: 1,883** $\rightarrow \sim 79$ **A100-hours—3.3 days on a *single* A100**, or ten hours on eight, against NucleicBERT’s 192.

The honest risk. Small could simply underperform. The evidence points the other way, but it is evidence about *related* tasks, not ours. The mitigation is that this is now cheap to test: train PHAROS-Small first and scale only if measurements demand it, with Base-v2 retained as the scale-up path. Combined with the staged token budget, the first informative checkpoint costs on the order of **one GPU-day**, so the sizing question becomes an experiment rather than an argument.

11 Dynamics: from a stiffness field to an ensemble

The brief asked for structure prediction “or maybe more to it, maybe dynamics of the RNA, maybe how it will fold”. This section says what the measured stiffness field licenses—and, as importantly, what it does not.

11.1 One structure is the wrong output for much of RNA

Riboswitches are *defined* by holding two functional folds, ligand-bound and unbound, so predicting one is predicting half the biology. The field has moved off the minimum-free-energy structure:

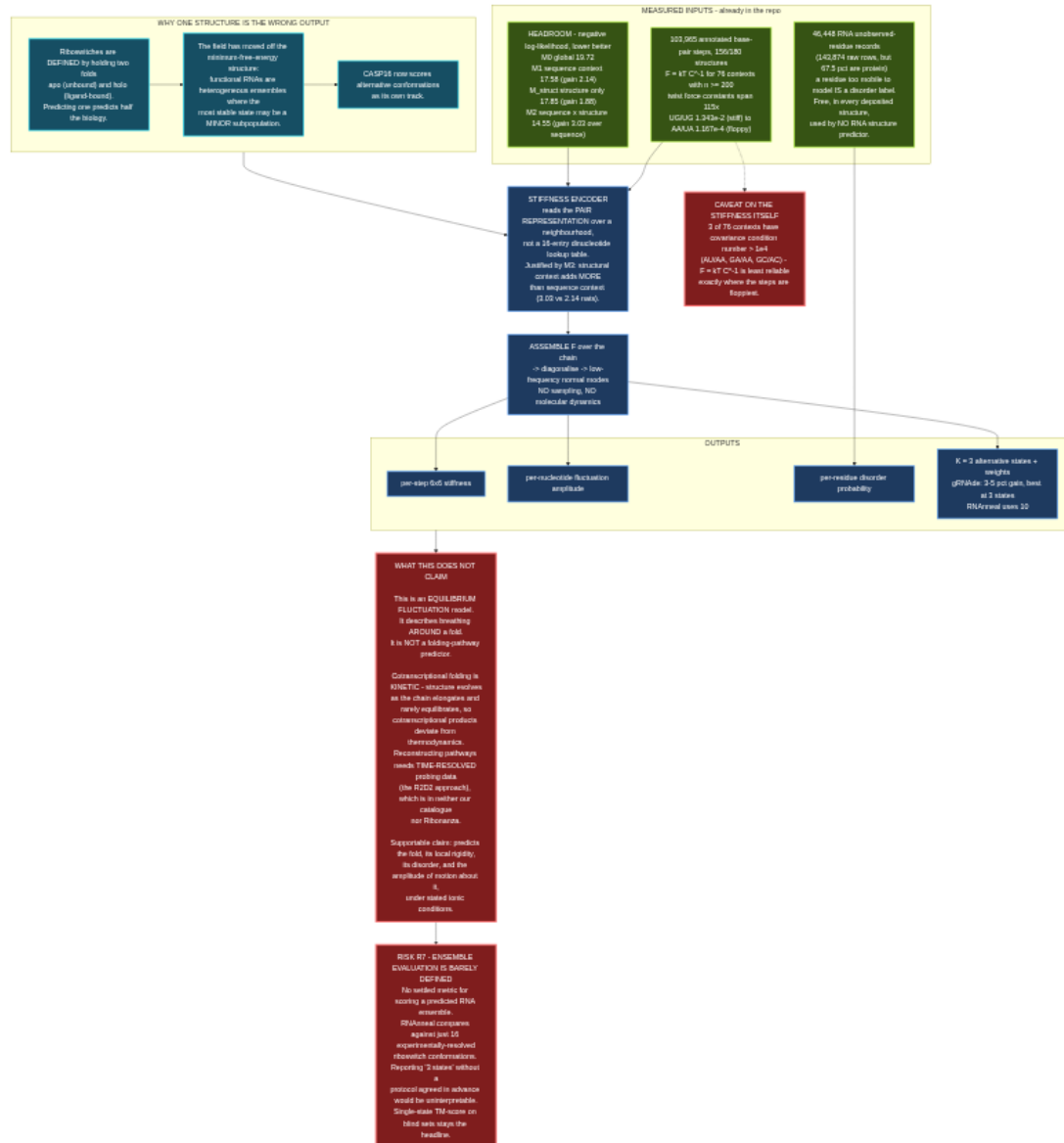


Figure 8: The dynamics module: measured inputs, the harmonic construction, and an explicit statement of what it does not claim.

functionally relevant RNAs are heterogeneous ensembles in which the most stable state may be a minor subpopulation, and CASP16 now scores alternative conformations as its own track.

11.2 The harmonic ensemble is nearly free

Given a predicted structure and a per-step 6×6 stiffness field, the harmonic fluctuation spectrum follows by diagonalising the assembled stiffness—no sampling, no molecular dynamics. This is the standard elastic-network construction, already assessed for RNA against MD and SHAPE. PHAROS therefore emits three outputs alongside coordinates.

Table 26: Dynamics outputs and their supervision. Two of the three label sources are already inside every deposited structure. **[measured]**

Output	Derived from	Supervision
per-step 6×6 stiffness	stiffness encoder	measured F , 76 contexts
per-nucleotide fluctuation amplitude	assembled $F \rightarrow$ normal modes	B-factors / RMSF
per-residue disorder	dedicated head	46,448 RNA unobserved-residue records
$K = 3$ alternative states	decoder samples, ranked	CASP16 alt-conformations

The disorder head is the cheapest addition in the design. A residue recorded in `_pdbx_unobs_or_zero_occ_residues` is one too mobile or disordered to model—a direct per-residue flexibility label, present in every deposited structure, used by no RNA structure predictor, and we extracted 46,448 RNA ones (of 143,874 rows across all polymers) at zero acquisition cost. On state count, gRNAd’s multi-state evidence is a consistent 3–5% improvement with **3 states** best, and RNAnneal uses 10; we default to $K = 3$.

11.3 Why the stiffness field is conditioned on structure

The literature is explicit that slide–rise, twist–roll and twist–slide couplings carry real physics, and that the **pentameric scale is the minimum range of elastic couplings**—concerted motion of neighbouring steps makes a dinucleotide step model demonstrably insufficient. Our own nested-model measurement reaches that conclusion independently.

Table 27: Negative log-likelihood of the step distribution under nested models; lower is better. **[measured]**

Model	NLL	gain
M0 global	19.7235	—
M1 sequence context	17.5792	2.14 nats over global
M _{struct} structural context only	17.8470	1.88 over global
M2 sequence $\times$ structure	14.5510	3.03 nats over sequence

Sequence and structural context are near-equal, complementary contributors (held-out +1.7847 and +1.7409; the combination adds a further +1.0336 beyond sequence). An earlier version claimed structure dominates; that was an in-sample artefact, corrected in Section 11 (REV-2).

11.4 What this does not claim

This is an equilibrium fluctuation model. It describes breathing *around* a fold. It is **not** a folding-pathway predictor. Cotranscriptional folding is kinetic—structure

evolves as the chain elongates and the system rarely reaches equilibrium, so cotranscriptional products routinely deviate from thermodynamic predictions. Reconstructing those pathways requires *time-resolved* probing data, which is in neither our catalogue nor Ribonanza. Describing a harmonic ensemble as “predicting how RNA folds” would be precisely the class of overclaim that the coevolution retraction in Section 3.4 was about. The supportable claim is: *predicts the fold, its local rigidity, its disorder, and the amplitude of motion about it, under stated ionic conditions.*

Risk this adds (R7). No settled metric exists for scoring a predicted RNA ensemble against experiment; RNAnneal compares against just 16 experimentally-resolved riboswitch conformations. Reporting “3 states” without a scoring protocol agreed in advance would be uninterpretable, so single-state TM-score on blind sets remains the headline and ensemble output is reported descriptively.

12 Generalization audit: can this handle *any* RNA?

The design claims contact maps, structure and other properties for RNA in general. This section tests that against the 180 sampled structures rather than asserting it. Four limits are real; three were undocumented before this audit.

Basis. Every figure below is re-derived under the single canonical RNA definition of §12: a residue is RNA iff its chain declares `_entity_poly.type = polyribonucleotide` and it occupies a polymer position, with hybrid DNA/RNA chains split per residue by the presence of an O2' atom. **179 of the 180 sampled structures carry countable RNA**, so every structure-count denominator here is 179. The exception is 7PU7, whose only nucleic entity is declared hybrid and modelled entirely as deoxyribonucleotide. Cycle 6 reported 162; that was defect #25.

12.1 G1 — single chains fit the context, whole entries do not

The longest RNA chain has median 70 nt and maximum 3,764 nt, so a 4,096 context covers every single chain we hold, with 56 of 179 structures whose longest chain exceeds 2,048. But *total* RNA per entry reaches **11,478** residues, and 44 of 179 entries exceed a 4,096 context. Outside this sample the gap is larger: lncRNAs (~ 19 knt) and viral genomes (~ 30 knt) are an order of magnitude beyond anything considered here. **Scope statement required:** PHAROS predicts single RNA chains up to 4,096 nt; it is not a whole-transcript or whole-ribosome model.

12.2 G2 — 99% of the structural evidence is RNA *in complex*

Table 28: How much of the 3D evidence base is isolated RNA. [measured]

Quantity	Value
Structures containing protein	158/179 = 88.3%
Median protein chains when present	10.5
Structures with more than one RNA chain	132/179 = 73.7%
RNA residues in protein-containing entries	98.95%

The architecture takes one RNA sequence and predicts its fold, but essentially all 3D supervision comes from RNA whose conformation is stabilised by partners the model never sees. Training on isolated chains asks the model to reproduce, from sequence alone, a fold that in reality exists only inside a ribonucleoprotein.

Note the asymmetry. **Twenty-one of 179 structures contain no protein at all, and they hold 3,257 residues — 1.05% of the corpus.** Isolated RNA is not rare as a *structure* count (11.7%); it is rare as a *residue* count, because isolated RNAs are small. Any “train on autonomous folds only” mitigation is therefore working with roughly one percent of the available supervision.

This is the single largest generalization hazard in the design, and it was not acknowledged anywhere before this audit. It is not fixable by tuning—it is a property of what RNA structural biology has deposited. The honest options are to restrict claims to plausibly autonomous folds (riboswitches, aptamers, ribozymes), to supply partner context as an input, or to report performance split by complexed versus isolated. The last is mandatory regardless.

12.3 G3 — the residue-weighted statistics are ribosome measurements

Sixty ribosome-like entries ($60/179 = 33.5\%$ of structures) contribute $286,458/309,197 = \mathbf{92.65\%}$ of all RNA residues. Every residue-weighted number in this report—stiffness, ion coordination, block occupancy, the contact-scaling law—is therefore predominantly a *ribosomal* measurement.

The mitigation is already present, by luck rather than design: the length-binned tables stratify it. Short chains show block occupancy 16.34% and effective c of 7.9 against 1.34% and 17.2 for long chains, so the $c = 20$ budget is generous at *both* ends and the sparse track is safe. But the headline “1.34% occupancy” is a ribosome figure and must be labelled as such rather than quoted as “RNA”.

Headroom warning. The 17.2 that sized $c = 20$ is a *mean* over long chains, and it was being read as though it bounded the maximum. Measured per chain across all 179 structures under both RNA definitions, the worst chain reaches an effective c of **19.04**: the budget has 4.8% **headroom**, not 16%. *Re-measured on 20,266 chains of the full corpus the maximum is 21.14 and $c = 20$ is breached; the budget is raised to 24.* Nothing in the 180-structure sample breaches it—0 of 115 chains checked, including all 24 with chains over 3,000 nt, where the worst is 18.44—so $c = 20$ is adequate *here*. It is adequate by a thin margin on a 180-structure sample, and the quantity that pushes a chain upward is the one the corpus under-represents: modified residues, which the ACGU filter had been deleting. Either widen c or measure the distribution on a larger sample before fixing it.

12.4 G7 — modified nucleotides cannot be tokenised

The vocabulary is five symbols. Measured canonically, **1.05%** of RNA polymer residues fall outside $\{A, C, G, U\}$ —3,237 instances across **82** distinct modification types, led by pseudouridine (1,003), OMG (373), A2M (365), OMC (274), OMU (270) and inosine (80). eLDORS is pre-normalised to five symbols, which is why vocab-5 looked sufficient for pretraining; structures are not normalised. Eighty-two chemically distinct modifications observed in 180 structures, led by the most abundant modification in cellular RNA, is a stronger case for widening the structural vocabulary than any percentage.

Re-derived in cycles 6–7 (defects #22–#25). Every analysis script had rolled its own definition of an RNA residue, and they disagreed on **72 of 180 structures**. mmCIF declares polymer type, so a residue is RNA iff its chain’s `_entity_poly.type` is `polyribonucleotide` and it occupies a polymer position; hybrid DNA/RNA chains are split per residue by the presence of an O2’ atom, a structural test rather than a name list.

G7’s published 8.90% was wrong by 8.5×: it counted DNA from hybrid duplexes (DT 4,608, DG 4,556, DA 4,440, DC 4,135) and UNK records (7,036) that are not in RNA chains at all—24,775 of the 27,437 “modified” instances. **G2 and G3 are unchanged from the published figures.**

Table 29: The G-findings under the canonical definition. The middle column is cycle 6, published here because it was wrong in an instructive way. **[measured]**

Finding	published	cycle 6	canonical
Structures with countable RNA	180	162	179
RNA residues	308,370	306,857	309,197
G1 longest chain, median	67	86.5	70
G2 structures with protein	88.3%	96.9%	88.3%
G2 more than one RNA chain	83.3%	72.8%	73.7%
G2 RNA residues in complexes	98.96%	99.70%	98.95%
G3 ribosome-like entries	33.9%	37.0%	33.5%
G3 residues ribosomal	93.07%	93.35%	92.65%
G7 outside $\{A, C, G, U\}$	8.90%	1.04%	1.05%
G7 distinct modification types	8	76	82

Defect #24 — partial propagation. The first pass at this correction updated four of the eleven affected numbers and stopped, and an unguarded global replace of the canonical values overwrote the *published* column of the comparison table in the architecture spec, making it read “99.70 → 99.70”. Three tables were left mixing the 180 and 162 denominators inside a single row-set. It is the third occurrence of the unguarded-replace failure mode, and the first to corrupt a table whose only purpose was to record a correction. Every substitution in the repair was made with an exact-match, count-asserted replace.

Defect #25 — half the format was never parsed. mmCIF serialises a category in two forms: a `loop_` form, and a key-value form (`_entity_poly.tag value`) used when the category has exactly one row. The resolver handled only the first, so **16 of 180 structures silently parsed to zero RNA**. They are not a random sixteen—a single polymer entity means a small isolated RNA, precisely the population G2 makes a claim about—and their absence inflated “RNA residues in complexes” from 98.95% to a spurious 99.70%. Cycle 6 reported that inflation as a finding: that *G2 strengthens*. **A defect introduced while auditing a definition produced a false result which the audit then published**, and it was caught only because the next cycle asked why 18 structures in a non-redundant *RNA* list appeared to contain no RNA. The resolver now parses both forms, and a test asserts that no structure resolves to an empty declaration.

Defect #26 — the same bug, in the other parser. With #25 fixed in the RNA resolver, the next question is whether any *other* parser shares it. The `loops()` helper in `extract_basepair_geometry.py`—used by the geometry, stiffness and ionic scripts—kept only the tag from each `_category.tag` line and discarded a value on the same line, so a key-value category yielded zero rows, silently, just as `_entity_poly` had. Measured: **9 rows across 9 structures**. The impact on published numbers is **+1 base-pair step** (103,964 → 103,965), **+1 RNA disorder record** (46,447 → 46,448) and one further structure with base-pair annotations (155 → 156); the 115.1× stiffness ratio, the A-form validation geometry and the 76 stiffness contexts are unchanged. Negligible in a 180-file sample—but the corpus is not 180 files, and the identical bug in `_entity_poly` moved a headline finding by 0.75 points. `_expt1` and `_expt1_crystal_grow` are key-value in *every* file that carries them and are read by a separate, key-value-aware parser; that was checked rather than assumed.

Defect #27 — the guard’s tolerance exceeded the effects it guards. Defect #26 was nearly missed because `verify_claims.py` reported OK for *annotated base-pair steps* = 103,965 against an expected 103,964: every claim defaulted to a 0.5% *relative* tolerance, and one part in 103,964 sits far inside it. Auditing all 54 numeric claims, 45 **carried a tolerance wider than the precision they quote**. The worst was *G7 fraction of RNA residues*, checked as

0.01047 ± 0.01 —it would have accepted anything from 0.0005 to 0.0205, a vacuous guard on the one figure this audit corrected by $8.5\times$. The default is now derived from how the value is written: an integer must match exactly, a decimal to half a unit in its last quoted place. All 54 still pass, which is the reassuring half: the claims were accurate to their stated digits, the guard was simply not proving it. Verified adversarially—injecting the retracted cycle-6 value (0.9970) and a +1 count drift both now fail the build, and both passed before.

Defect #28 — the cross-document check proved the wrong thing. The same guard verifies that each headline appears in all three deliverables, using substring containment. That proves a string occurs *somewhere*, not that it occurs in the claim it guards. Of 43 guarded tokens, **seven are three characters or fewer and match inside longer numbers**: half of 61’s matches in the architecture spec sit inside other numbers, and its first standalone match is a contact-sparsity table row, nothing to do with the 61M active-parameter claim. Measured adversarially by deleting *every* occurrence of the 149M/61M/153M/A100-hour/115 $\times$ claims: the old check **still passed three of the five**. Numbers are now matched at number boundaries, with explicit context patterns for the seven ambiguous tokens, and all five fail when the claim is removed. Tightening also exposed two *latent* defects—the tokens 15.342 and 16.376 were written truncated and had only ever matched as prefixes of 15.3424 and 16.3760, so they had never guarded those values exactly.

C14 — does the definition change the residue-weighted numbers? Cycles 6–7 fixed the definition and re-derived only the G-findings. Every other residue-weighted number still came from a script filtering on `label_comp_id` $\in \{A, C, G, U\}$, which does not merely narrow the label set—it *deletes* modified residues from the middle of a chain, so the neighbours become adjacent and chain length, sequence separation, the SEQ_SEP guard and the block indices $[i/b]$ all shift.

Table 30: Sensitivity of the residue-weighted measurements to the RNA definition, measured chain-by-chain over all 179 structures. **[measured]**

Quantity	published (ACGU)	canonical
Chains whose length changes	—	31/91 = 34.1%
Residues restored	—	703
Median L , affected chains	—	+0.87%
Median contacts/nt, affected	—	+1.37%
Median effective c , affected	—	+1.19%
Worst single chain (7VNV)	$L = 61$	$L = \mathbf{78}$ (+27.9%)
Worst contacts/nt	—	+46.1%
Max effective c	19.03	19.04
<i>full corpus, 20,266 chains</i>	—	21.14
Chains breaching $c = 20$	0	1 (<i>full corpus</i>)

The 24 entries with chains longer than 3,000 nt were checked separately rather than excluded: max effective c is 18.44 under *both* definitions.

The corpus statistics and the $c = 20$ sizing decision both survive—the medians move by about one percent and nothing breaches the budget. **The per-chain error does not.** A 27.9% length error is invisible in a median and unacceptable as a model input, so the published statistics stand while the training pipeline must use the shared canonical loader `mmcif_entities.longest_rna_chain()`. That distinction—corpus statistics versus per-example correctness—is the substantive result here.

12.5 G6 — router behaviour at recycle 0 is undefined

The router is specified to read pairing probability, predicted rigidity and local density—all produced by the pair track, which runs *after* the trunk. At recycle 0 none of them exist. B_{elec} has exactly this problem and it was resolved explicitly; the router never was. The fix has the same shape—route on N_{eff}/L and sequence-local features on the first pass, switch to structural routing from recycle 1—but it has to be stated, because an undefined router decides which experts fire.

12.6 Verdict

Efficiency is the best-verified part of the design. The hierarchical pair track runs at 0.96% of dense at $L = 4096$ while the dense baseline cannot run at all; the parameter and FLOP arithmetic reproduces exactly; training is costed at ~ 79 A100-hours against NucleicBERT’s 192 GPUs.

Generalization to “any RNA” is not supportable as scoped. Four measured limits and one undefined behaviour bound it. None invalidates the design; together they fix the honest headline: *a single-chain RNA structure model, up to 4,096 nt, trained predominantly on ribosomal and complex-embedded RNA, which must report performance split by isolated versus complexed context.*

13 Decided parameters and formats

13.1 A conflict the design could not satisfy

Three requirements were mutually incompatible: the *smallest viable model* (61M active), the *full 323B-token corpus*, and *4-bit* precision. *Low-Bit Quantization Favors Undertrained LLMs* (ACL 2025, arXiv 2411.17691, 1500+ checkpoints) finds quantization-induced degradation is worst for small models trained on many tokens, stating that “smaller models require higher precision at high training token counts”.

Table 31: Where each size sits relative to that law. **[measured]**

Model	active	tok/param @323B	× Chinchilla	4-bit
Micro	12M	26,917	1,346×	worst case
Mini	30M	10,767	538×	worst case
Small	61M	5,295	265×	worst case
Base-v2	269M	1,201	60×	risky

RNA worsens it: measured sequence entropy is 2.0167 bits/nt against roughly 11 for an English token, so the corpus is more redundant than its token count implies.

13.2 Decisions, with three reversals

D1 keeps PHAROS-Small. **D2** (reversal) cuts the token budget from 323B to a staged 25B, stopping on plateau—the over-training defence was argued at 269M active, and at 61M it becomes 265× Chinchilla while buying no information. **D3** (reversal) trains in BF16, adopts FP8 on Hopper once the baseline reproduces, and takes *no* 4-bit at this size. **D4** defers QAT as a deployment technique. **D5** routes new attributes in as input features through one projection rather than a wider d_{model} . **D6** adopts two vocabularies. **D7** (reversal) makes the cost hardware-explicit.

Why not 4-bit here. NF4 cannot train from scratch at all—it is W4A16 with a *frozen* 4-bit base, gradients reaching only LoRA adapters. NVFP4 *is* a genuine pretraining format, reporting no measurable loss against an FP8 baseline, but its validation point is 8B parameters and 1T

tokens (125 tok/param) against our 5,295. MXFP4 needs $\sim 36\%$ more tokens than NVFP4 for equal loss.

Defect #15: the published cost was not a real unit. A100 (Ampere) has no FP8 tensor cores; FP8 arrived with Hopper. The cost was quoted in **A100-hours** while a $1.60\times$ **FP8** lever was applied, so “ ~ 79 A100-hours” existed on no single machine and “3.3 days on a single A100” was unreachable. Corrected, with levers split by what they depend on (hardware-independent: Muon $2.00\times$, read down-weighting $1.19\times$, combined $2.38\times$): at 323B tokens, 126 h on A100 bf16 or 20 h on H100 fp8; at the decided 25B budget, ~ 10 h and ~ 1.6 h respectively. **The token cut is a $12.9\times$ saving against FP8’s $1.6\times$, and unlike a precision lever it costs no accuracy because the dropped tokens are redundant.**

Where the over-provisioned bits go. A fully attributed nucleotide carries 58.9 measured bits; a 512-dim bf16 token holds 8,192— $139\times$ over-provisioned. d_{model} is *compute* space, not storage, so attributes are nearly free while width is not: FLOPs go as $6N_{\text{active}}$ and widening d is quadratic in the FFN. Only 26.0 of those bits are available at inference; the remainder are supervision targets, and feeding them in would leak the answer.

14 Defect #17: refinement loops were never costed

The cost model computed $6N_{\text{active}}T$ — the cost of *one* pass through the blocks. PHAROS-Small runs **8 refinement loops**, and the training design specifies deep supervision at *every* segment with a one-step (detached) gradient, so each segment is its own forward and backward pass. Compute therefore scales linearly with the loop count. The design even states that loops “cost compute but no additional activation memory”, and then omits the compute. Activation memory indeed does not scale—that is what the detach buys—but FLOPs do.

Table 32: The ladder with the loop factor restored. Every published cost was understated by exactly its loop count. **[measured]**

Model	blk	loops	train d.	serve@3	active	eff. comp.	A100-h
Small	16	8	128	48	61M	488M	78
Base-v2	32	3	96	96	269M	807M	129
Mini	12	12	144	36	30M	360M	58
Micro	8	16	128	24	12M	192M	31

Two headline claims do not survive.

“ $4.4\times$ smaller than Base-v2” collapses to $1.65\times$. By active parameters, 61M against 269M is $4.4\times$. By *effective compute* (active $\times$ loops) it is 488M against 807M — only $1.65\times$. Eight loops against three consume most of the saving.

The depth comparison mixed training with serving. Small’s 8 is a *training* loop count (the design says “train 8–16, serve 3”), while Base-v2’s 3 reads as a serving count—two different quantities in one column. At a matched serve-3 setting, Small is **48** effective layers against Base-v2’s **96**: **the claim reverses at inference.**

Corrected cost. The lever chain falls from a claimed $24\times$ to $5.6\times$, because the right-sizing step is $1.65\times$ rather than $4.43\times$ once loops are counted. At the decided 25B-token budget, PHAROS-Small costs **78** A100-hours in bf16 or **12** on H100 fp8. Two independently written cost paths agree on both figures.

The token-budget cut ($12.9\times$) therefore remains the largest single lever in the design—larger than right-sizing and larger than any precision choice.

Two spec gaps closed in cycle 4. The attention **head count was never specified anywhere** ($d = 512$, heads unstated); it is now fixed at 8 heads of 64 dims. And “Motif bank + heads + decoder $\sim 10\text{M}$ ” was a placeholder whose **decoder was never specified at all**, so the line could not be checked. Itemised, the 9 heads cost 1.65M—the three added for the dynamics module only 0.80M between them—the frozen motif bank 0.17M, and a plausible 4-block frame-diffusion decoder **12.59M**, which alone exceeds the line. Totals rise from 149M/61M to $\sim 153\text{M}/\sim 65\text{M}$. Heads are cheap; the decoder is not.

15 Buildability audit

Nineteen defects in, the newest failure mode was *components never specified at all*—the attention head count and the decoder size—then silently assumed small. This section asks one question of every component: could a competent engineer implement it from the specification alone, without inventing a number?

Thirty-seven components were enumerated; ten were not specified. Seven were genuinely absent (learning rate, batch size, warmup, the five loss weights, diffusion steps, the MoE bias-update rate, the Gated DeltaNet state size); two existed only in the reference code or the data catalogue rather than in any config; one exposed a circularity. All are now closed or marked PROVISIONAL with the procedure that fixes them. Invention would have been worse than deferral: the 5B checkpoint costs about 1.5% of the run, so each is a cheap sweep rather than a guess.

15.1 Defect #20: B_{motif} shares the router’s circularity

Three bias terms consume output produced *after* the trunk they feed, not one. B_{elec} needs a distance estimate and was given an explicit first-pass rule. B_{motif} needs the motif bank’s key, which is the *interaction graph* produced by the pair track—and never had one. The router’s own fix was described but never written into the bias equation. All three now share a single explicit recycle schedule in the model config.

15.2 Defect #21: the MoE copied one DeepSeek ratio and not the other

The design cites DeepSeekMoE fine-grained segmentation as its template. Checked against the verified DeepSeek-V3 configuration ($d = 7168$, expert $d_{\text{ff}} = 2048$, 1 shared plus 256 routed, 8 activated):

Table 33: Per-expert width matches the template; active capacity does not. **[measured]**

	d_{ff}/d	active experts	active FFN / d
DeepSeek-V3	0.286	9	2.571 $\times$
PHAROS-Small	0.250	6	1.500 $\times$
dense FFN	—	—	4.000 $\times$

PHAROS’s active FFN capacity is $1.7\times$ below the template it cites and $2.7\times$ below a dense FFN. The design copied the segmentation and not the activation, and the ratio had never been examined. This is a design gap rather than an error—PHAROS is deliberately data-limited, so less active capacity may be correct—but it is exactly the kind of parameter that silently caps quality. The fix is unusually cheap: top-8 of 32 gives $2.50\times$ for +12.6M active parameters and essentially no change in total, because the experts already exist. The decision is deferred to the 5B checkpoint.

16 Novelty claims

Table 34: Novelty claims stated precisely, with prior-art status.

Claim	Prior-art status
Ionic conditions as a model input, reaching attention logits through a Manning-screened Coulomb bias	No existing RNA structure predictor accepts ionic conditions. The physics term needs no labels; the learned response requires titration data (Section 5).
First sparse-MoE architecture for RNA structure	AIDO.Protein is MoE for protein; no RNA equivalent exists.
Hierarchical coarse-to-fine pair track selecting blocks rather than pairs, justified by measured 1.34% block occupancy	AlphaFold3 and Rhoformer are dense $O(L^2)$; flat top- K sparsification is measured here to fail at $\sim 20\%$ recall on long chains.
Frozen motif KV bank as a retrieval-based geometry prior	The Motif Atlas is published but wired into no large model.
Mg ²⁺ sites and B-factor rigidity as free auxiliary supervision	Present in mmCIF; used by no structure predictor.
Coupled ion-rigidity expert justified by a measured 1.523σ gradient	Treated separately or not at all elsewhere. Partial correlation controlling for packing density is $+0.372$ vs $+0.420$ raw.
Learned stiffness encoder emitting a per-step 6×6 precision matrix trained by Gaussian NLL, replacing a tabulated force field	Nucleic-acid elasticity uses fixed per-context stiffness tables; measured here to capture under half the signal (14.551 vs 17.579 nats/step).
Physics labels mined from unused mmCIF categories: 103,965 step geometries, 44,708 curated Mg ²⁺ coordinations, 143,874 disorder records	These categories ship with every RNA structure and are used by no structure predictor.
Depth-gated coevolution routing, with N_{eff}/L as a router feature	RhoFold+ concatenates language-model and MSA features at fixed weight; no method routes on measured alignment depth. Motivated by the literature and by measured between-family variance (0.333 – 1.000); our own depth split is weak evidence (Spearman $+0.224$, $n = 12$).

17 Risks and open questions

1. **Block-detection recall.** Superseded twice, still not eliminated. The flat top- K design was measured at $\sim 20\%$ recall and replaced (Section 4.3). Testing then showed the replacement’s selectors received *no gradient at all* and could never have been trained; that is now fixed by score gating plus a directly supervised block-occupancy loss. The track has a 100% recall ceiling and the selectors now have labels, but no model has been trained, so attainment remains unproven.
2. **Ionic metadata sparsity—audited, partially confirmed** (Section 5). About 36% of structures yield a recoverable ionic condition. Worse, recorded Mg²⁺ spans only 5–15 mM because the PDB is survivorship-biased. The learned ionic response must come from RMDB titration series, not the PDB. The closed-form physics term and the Mg²⁺ site head are unaffected.

3. **Crystallographic ions are not the solution ensemble.** The explicit head learns site binding, not the diffuse atmosphere. These must not be conflated.
4. **B-factor is a noisy rigidity proxy.** Excluding cryo-EM left only 31 of 137 parsed structures usable, which materially shrinks the rigidity label set.
5. **Association, not isolated causation.** Mg^{2+} proximity and packing density both track folded cores.
6. **Infrastructure.** NucleicBERT used 192 A100s densely. MoE improves FLOP efficiency but adds routing and communication complexity.
7. **Evaluation honesty.** The field ceiling is $\text{TM} \approx 0.55$. Any improvement claim must be made on blind sets—CASP15/16 and RNA-Puzzles—never on rRNA-saturated random splits. Sequence-identity deduplication is mandatory.

18 Scaling the evidence: the corpus and raw-archive passes

Everything up to this point was measured on 180 non-redundant structures fetched onto a laptop. Two later passes re-derived those measurements at scale: a *corpus* pass over 29,807 chains of RNA3DB and RNASolo, and a *raw-archive* pass over the 10,527 PDB entries that contain an RNA polymer entity, acquired in full (15 GB) because three of the claims describe entries rather than chains and are therefore not computable on per-chain extracts at all. Of those, **10,520** hold an RNA polymer chain: 29,038 chains, 13,348,166 residues, $43\times$ the 309,197 residues §18 supersedes.

This section states what changed. It does not revise the sections above: those record the $n = 180$ analysis and its eleven audit cycles, and the value of that record is that it is what was actually believed at the time.

18.1 Every mean held; every maximum failed

	$n = 180$	full corpus	
<i>Central estimates</i>			
mean effective c , long chains	17.2	17.14	holds
sequence entropy (bits/nt)	2.0167	2.0165	holds, 238M nt
$b=4$ block occupancy	1.34%	1.67%	holds
contact separation, band 512	0.812	0.8545	holds
modified-residue fraction	1.05%	1.005%	holds, 11.6M res
Mg-rigidity gradient	1.76σ	1.523σ	holds, -13%
<i>Extreme quantiles</i>			
max effective c	19.04	21.14	breach
longest chain	3,764	4,450	breach
chains over 4,096	0	8	breach
max contacts/nt	5.50	7.66	breach
Mg:K ratio	9:1	54:1	wrong $6\times$

This is not eleven separate errors. It is one property of an $n = 180$ sample, which estimates a mean well and cannot estimate a $p_{99.9}$ at all. Two load-bearing parameters—the pair-track budget `target_c` and the context window—had been set from maxima. The rule adopted in response: **no parameter may be set from an extreme quantile measured on fewer than 10^3 structures**; where the corpus cannot supply that, the parameter is set from a mean plus a stated safety factor.

	$n = 180$	raw PDB	
G2 residues in entries with protein	98.95%	97.15%	holds, -1.8 pts
G2 isolated RNA, share of residues	1.05%	2.83%	$2.7\times$ larger
G3 residues from ribosome-like entries	92.65%	85.94%	holds, -6.7 pts
G3 ribosome-like entries	33.5%	20.88%	-12.6 pts
G1 longest single RNA chain	3,764	4,450	reproduces exactly
G1 entries over 4,096 RNA residues	24.6%	15.06%	over-stated
G1 total RNA per entry, max	11,478	22,345	$1.95\times$ breach

18.2 The raw archive closes G1, G2 and G3

Both aggregate claims held, and both moved *downward*: the archive is less protein-bound and less ribosome-dominated than BGSU representative sets—which deliberately over-weight large assemblies—made it look. Every residue-weighted figure inherited from the $n = 180$ analysis is therefore pessimistic about diversity rather than optimistic. That is the safe direction to have been wrong in, and it is still wrong.

The consequence that matters is not the headline but the stratum underneath it. G2 mandates that evaluation be reported split by isolated versus in-complex RNA. The isolated stratum is 2,350 entries and 378 thousand residues, $2.7\times$ what $n = 180$ measured—large enough to *train* on, not merely large enough to report a number for.

18.3 One maximum stopped moving

Scanned across every RNA polymer chain in all 10,520 RNA-bearing entries, the longest single RNA chain is **4,450** nt (6HRM), and exactly five entries exceed 4,096: 6HRM 4,450, 7UPH 4,438, 4V6X 4,298, 8TOC 4,269, 7LHD 4,217, all large ribosomal rRNA. These are the same five structures, at the same lengths, that the derivative corpora found.

A maximum that does not move when the sample grows from 29,807 derived chains to the entire archive is not an estimate any more. It is the value, over a closed population, and there is no larger sample to be had. The context window is therefore set to **4,608**: it covers every RNA chain that has ever been solved, with 158 nt of headroom. Choosing 4,096 to save 11% of the position budget would leave the five largest rRNA chains—among the hardest and most-studied folds in the corpus—permanently outside the model.

18.4 One maximum did not: `target_c`

The pair-track budget was raised $20 \rightarrow 24$ provisionally, because the 21.14 that forced the raise came from corpora carrying 0.025% modified residues against raw PDB’s 1.005%—a $40\times$ under-representation of precisely the quantity that drives the tail. Re-measured on 14,106 chains from the raw archive, at 0.476% modified residues:

	$n = 180$	derivatives	raw
chains measured	180	20,266	14,106
max effective c	19.04	21.14	23.30
$p_{99.9}$	—	19.52	23.08
chains over 20	0	1	34
chains over 24	—	—	0

The maximum did rise and 24 still contains it: zero of 14,106 chains breach the budget across the complete archive. But the stated mechanism was wrong, and the right one generalises further than `target_c` does.

Splitting the raw chains by whether their entry appears in RNA3DB or gRNAdel/RNASolo: the 12,573 chains from *covered* entries reach **21.14**—*exactly* the published derivative figure—and the

1,533 chains from uncovered entries reach **23.30**. The derivative measurement was not distorted by missing modifications; it was correct for the population it contained. The entire difference is **coverage**.

More precisely: the top 30 chains in the whole archive are a single deposition campaign, and all 30 are entries no derivative holds—the 9T-series *E. coli* ribosome peptidyl-transferase-centre focused refinements, ~596-nt fragments of 23S rRNA at 2.28 Å in rRNA-methyltransferase-knockout strains. 9T1E was deposited **2025-10-21**, after RNASolo’s 2023-11 snapshot. The first chain belonging to any other structure is **rank 31**, at 21.14.

Modification enrichment is real but secondary: the top 34 carry 1.98% modified residues against the corpus’s 0.48%, but against a *length-matched* control (500–700 nt chains) the enrichment is only **1.9×**, not 40×. These are methylation-variant ribosome structures and are modification-rich by design—which describes what happens to sit at the top, not why the derivatives missed it.

The lesson is not about modified nucleotides. A parameter fitted to a curated derivative inherits that derivative’s snapshot date, and the extremes live in precisely what it omits.

But the headroom is **2.9%**, not the 12% originally claimed, against a statistic that has risen at every single re-measurement—19.04 → 21.14 → 23.30, roughly +2.1 per pass. Unlike the chain-length maximum, this is not a closed population; it is a property of whatever is deposited next. **The overflow path is therefore load-bearing rather than defensive:** the track truncates to budget, records the miss, and the recorded miss-rate is a training metric, not a debug counter. Raising to 28 would buy headroom for 34 chains in 14,106 and cost proportionally in L3 refinement on all of them.

The effective sample size at the top of this distribution is 1, not 14,106. Ranks 1–30 are the same ~596-nt molecule solved thirty times in one campaign. Ranks 31–34 are 7PAS (21.14), 9Z80 (20.97—a 119-nt chain in which *all* 119 residues are modified), and 4V6O/4V6P (20.11/20.01). **Four independent molecules exceed 20 in the entire Protein Data Bank.** A budget validated against that tail is validated against very little, which is the argument for the overflow path and against reading 23.30 as a ceiling.

18.5 Two defects, both the same shape

The raw pass found two, and both are the recurrence of an earlier one—a second definition of something the canonical resolver already defines, drifting from it.

C15. `entity_poly_types()` returns *auth* chain identifiers. Two analysis scripts had each grown a private entry counter that joined to it on `label_asym_id`. Where an entry’s two labellings differ—overwhelmingly older depositions, e.g. 1ARJ, whose sole RNA chain is `label A / auth N`—every atom row missed its entity and the entry reported zero polymer residues of any kind. It affected **3,254** of 10,527 entries, 686 of them reporting no polymer at all, and it deflated G2 to 89.2% and G3 to 33.6% before being caught. What caught it was not a test: the scan reported thousands of RNA-free entries in a corpus selected for containing RNA, and 1ARJ—a textbook TAR RNA structure—was on the list. The number was implausible on its face.

C16. The geometry resolver never read `pdbx_PDB_model_num`. An NMR entry deposits an ensemble—20 models by convention—and every model repeats every atom, so all 20 collapsed into one residue: 1ARJ returned **424** heavy atoms per residue against a nucleotide’s real ~21. Geometry built from that is the *union* of the ensemble’s contacts, so two residues count as touching if they touch in any single model, and both the contact count and effective *c* inflate for every NMR chain. Blast radius was checked rather than assumed: the four chains setting the published maxima (7PAS, 6Q95, 1VY7, 6XHV) are single-model cryo-EM or X-ray, and sampling 400 files from each derivative corpus found zero multi-model files, so no published statistic moves. The defect only ever bit the raw-entry path, which is new.

Both are fixed, both are asserted by tests, and counting now lives once—beside the resolver whose keys it has to match.

18.6 A quarter of the archive was outside the training corpus

Matching the entry identifiers of both derivative corpora against every RNA-bearing entry in the archive:

	entries
RNA3DB	5,389
gRNAd / RNASolo (2023-11 snapshot)	6,156
union of the two	7,943
RNA-bearing entries in the PDB	10,520
in neither derivative	2,581 (24.5%)

All 2,581 are already on disk. Their median longest chain is 22 nt—the uncovered population is dominated by small crystallographic oligos, which is presumably why the derivative pipelines dropped them—but the tail is not small, and it points at the three things this corpus is shortest of. **608** entries carry a chain inside the 64–3,000 training window that neither derivative exposes. **925** are protein-free, which is exactly the isolated-RNA stratum G2’s mitigation is evaluated on. And, as the previous subsection showed, this set contains the entire top 30 of the effective-*c* distribution—so it is not a marginal 24.5% but the part of the archive where the parameters are decided.

19 Status

This is a design specification. No model has been trained and no predictive performance is claimed. What has been established is the evidentiary basis: the contact-scaling law, the ion–rigidity coupling, the ion inventory and coordination chemistry, and the GNRA negative result—measured first on 180 non-redundant structures, then re-derived on 29,807 chains and on the 10,520 RNA-bearing entries of the entire PDB (§18). Every central estimate reproduced; five of six extreme quantiles did not, and the one that did—the 4,450 nt longest chain—did so over a closed population and is therefore settled rather than merely confirmed. 214 automated checks pin every quoted number to the JSON artefact it came from and fail the build on drift.

The next phase is to **train the block-detection scorer**. The reference implementation proves the sparse pair track’s cost and structure; it does not prove that a model can find the occupied blocks, and that is now the largest unvalidated assumption in the design by a clear margin.

A Reproducing the measurements

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Script	Output
.../sampling/fetch_samples.py	180 mmCIF, 12 Rfam seeds
.../sampling/analyze_ions_motifs.py	ion_summary.json
.../sampling/analyze_rigidity.py	rigidity_summary.json
.../sampling/analyze_contact_sparsity.py	contact_sparsity.json
.../sampling/validate_proposal_recall.py	proposal_recall.json
.../sampling/analyze_contact_separation.py	contact_separation.json
.../sampling/analyze_block_sparsity.py	block_sparsity.json
.../sampling/audit_ionic_metadata.py	ionic_metadata_audit.json
.../sampling/audit_em_buffers.py	em_buffer_audit.json
.../sampling/test_residue_guard_bias.py	residue_guard_bias.json
.../sampling/audit_generalization.py	generalization_audit.json
.../sampling/precision_and_hardware.py	precision_hardware.json
.../sampling/audit_token_attributes.py	token_attributes.json
.../sampling/decide_precision_and_size.py	precision_size_decision.json
.../sampling/recompute_cost_with_loops.py	cost_with_loops.json
.../sampling/extract_basepair_geometry.py	basepair_geometry.json
.../sampling/measure_stiffness_headroom.py	stiffness_headroom.json
.../sampling/measure_coevolution.py	coevolution_signal.json
.../reference/benchmark_pair_track.py	pair_track_benchmark.json
.../sampling/extract_basepair_geometry.py	basepair_geometry.json
.../sampling/measure_stiffness_headroom.py	stiffness_headroom.json
.../sampling/verify_claims.py	regression guard (exit 0 = reproduces)
.../reference/test_hierarchical_pair_track.py	correctness tests (exit 0 = all pass)
.../sampling/training_cost_model.py	training_cost_base.json
.../sampling/optimize_architecture.py	architecture_optimization.json
.../sampling/parameter_budget_justification.py	sizing analysis (stdout)

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